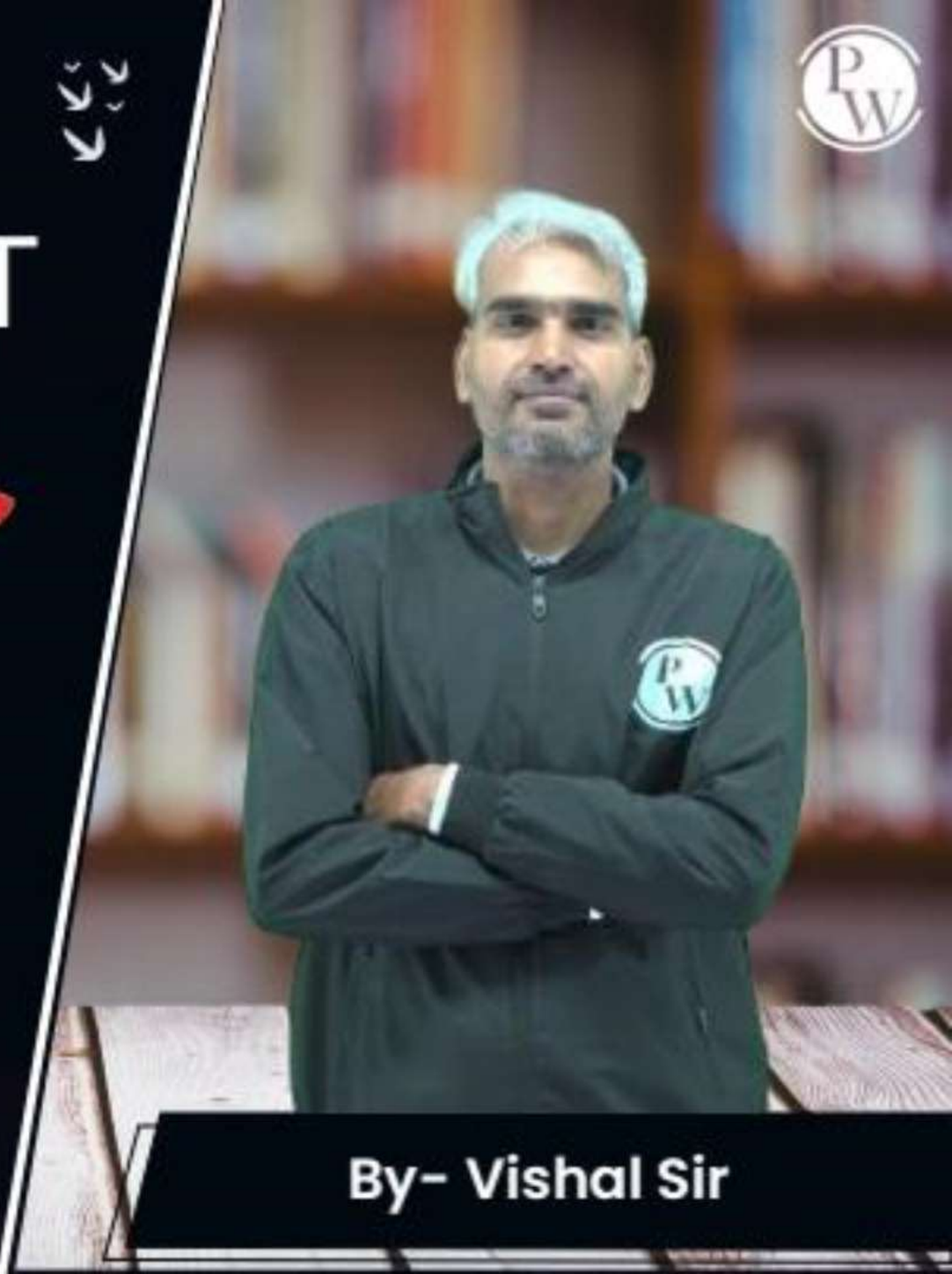
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 14



By- Vishal Sir

Recap of Previous Lecture



Topic Sublattice

Topic Bounded lattice

Topic Complements of an element in a lattcie

Slide

Topics to be Covered



- Topic: Complemented lattice
- Topic: Distributive lattice
- Topic: Boolean lattice / Boolean algebra

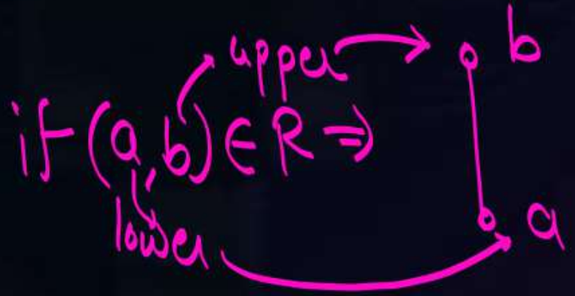
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Topic : Hasse Diagram / POSET Diagram

In a Hasse diagram of a POSET, (A, R)

1. There is a vertex corresponding to every element of set.
2. There is an edge from vertex a to vertex b only if a relates to b and there is no element x in the set such that a relates to x, and x relates to b.(Transitivity is implied in the Hasse diagram not represented explicitly)
3. No self-loop on the vertices (i.e. reflexivity is implied in the Hasse diagram not represented explicitly).
4. It is not directed but it uses implied upward orientation.



Slide



Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSETs

1) $(\{-1,0,2.5,4,6\},\leq)$
2) $(D_6,\div)$
3) $(D_{12},\div)$
4) $(\{2,3,4,6\},\div)$
5) $(\{2,3,6,12\},\div)$
6) $(\{1,2,3,4,6,9\},\div)$



Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{-1,0,2.5,4,6\}, \leq)$

It is a totally ordered set



$(\{-1,0,2.5,4,6\}, \leq)$ is a totally ordered set.

Hasse diagram for a TOSET will always look like a linear chain

∴ Totally ordered set are also known as "Linearly ordered set"

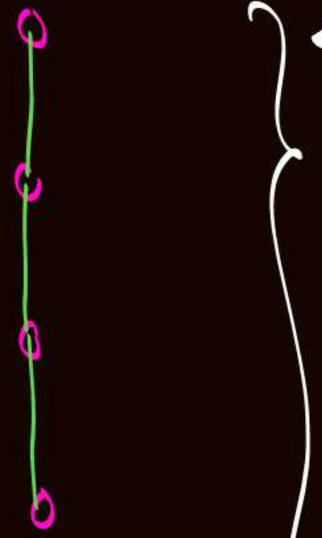
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Q: Let $A = \{a, b, c, d\}$

How many total orders (total order relation) are possible on set A.

Soln:- Hasse diagram of a totally ordered set will always look like a linear chain.

- Hasse diagram for a TOSET with 4 elements will look like



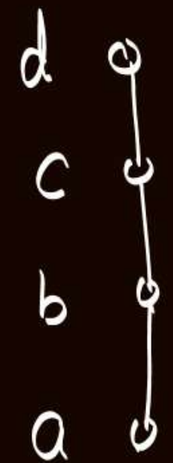
← This 4 vertices are w.r.t. 4 elements of set.

We can place 4 elements of the set in 4! ways, in Hasse diagram of 4 vertices

∴ 4! total order relation are possible

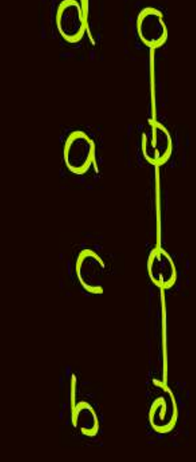
Total order

① Let w.r.t total order $:= R^1$



$\Rightarrow$ $R_1 = \{(a,a), (a,b), (a,c), (a,d), (b,b), (b,c), (b,d), (c,c), (c,d), (d,d)\}$

② Hasse diagram w.r.t. total order R_2



$\Rightarrow$ $R_2 = \{(b,b), (b,c), (b,a), (b,d), (c,c), (c,a), (c,d), (a,a), (a,d), (d,d)\}$

Similarly 4! different total order relation are Possible on set $A = \{a,b,c,d\}$

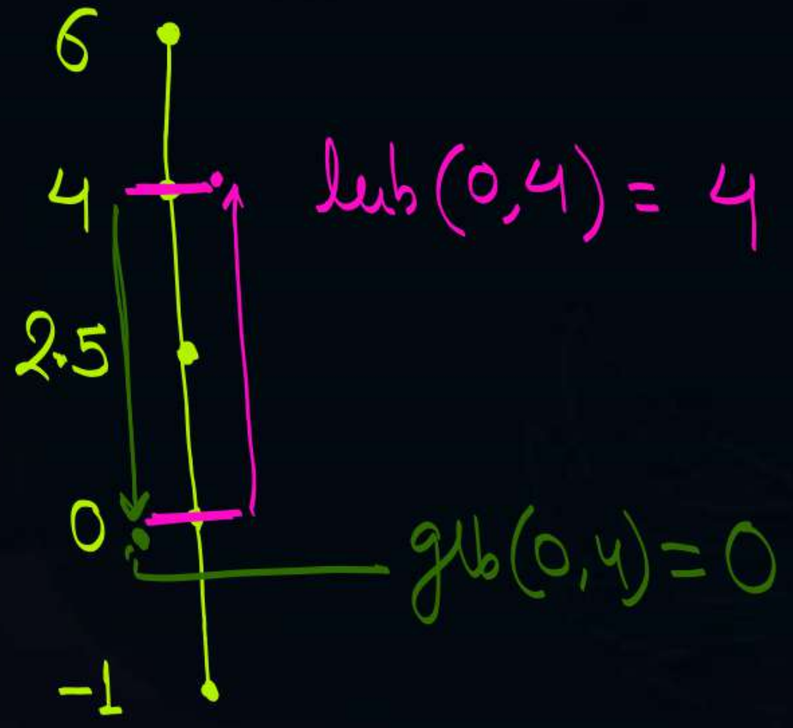
Note: If A is any set of cardinality 'n', then $n!$ total order relation are possible on set A.



Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{-1,0,2.5,4,6\}, \leq)$



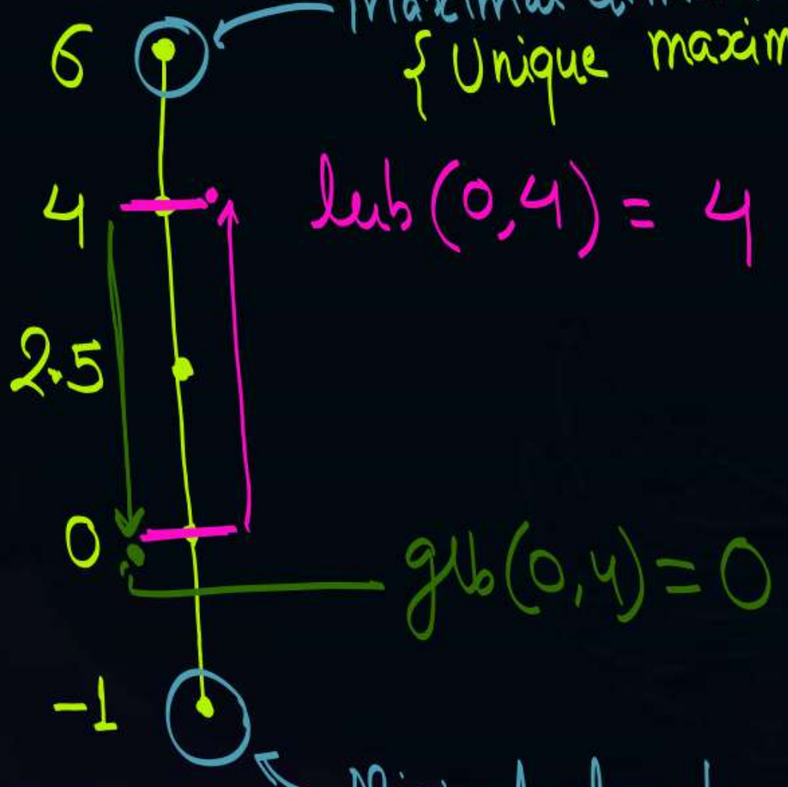
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Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{-1,0,2.5,4,6\}, \leq)$

lub as well as glb exist for every pair of elements.
∴ It is a lattice
i.e. Join semi lattice as well as Meet semi lattice



Maximal element { Unique maximal, ∴ also maximum element }

lub(0,4) = 4

glb(0,4) = 0

Minimal element { Unique Minimal. ∴ It is minimum element as well }

Slide



Topic : Hasse Diagram / POSET Diagram

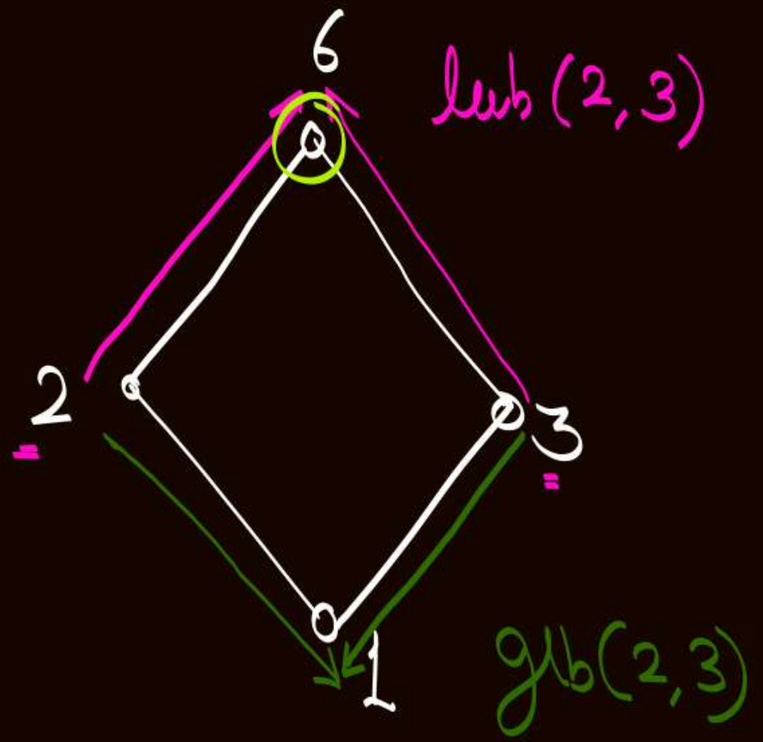
Draw the hasse diagram for the following POSET

$(D6, \div)$

Slide

$(D_6, \div)$

$D_6 = \{1, 2, 3, 6\}$



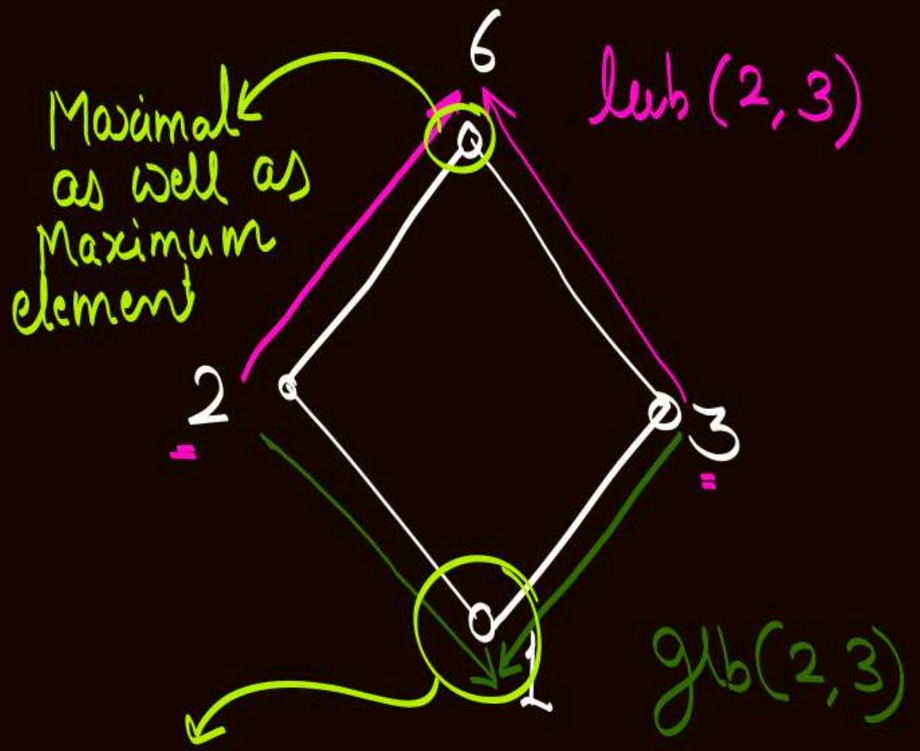
$(D_6, \div)$

$D_6 = \{1, 2, 3, 6\}$

lub as well as glb exist for every pair of elements,

$\therefore$ It is a lattice

- i.e. Join semi lattice as well as Meet semi lattice





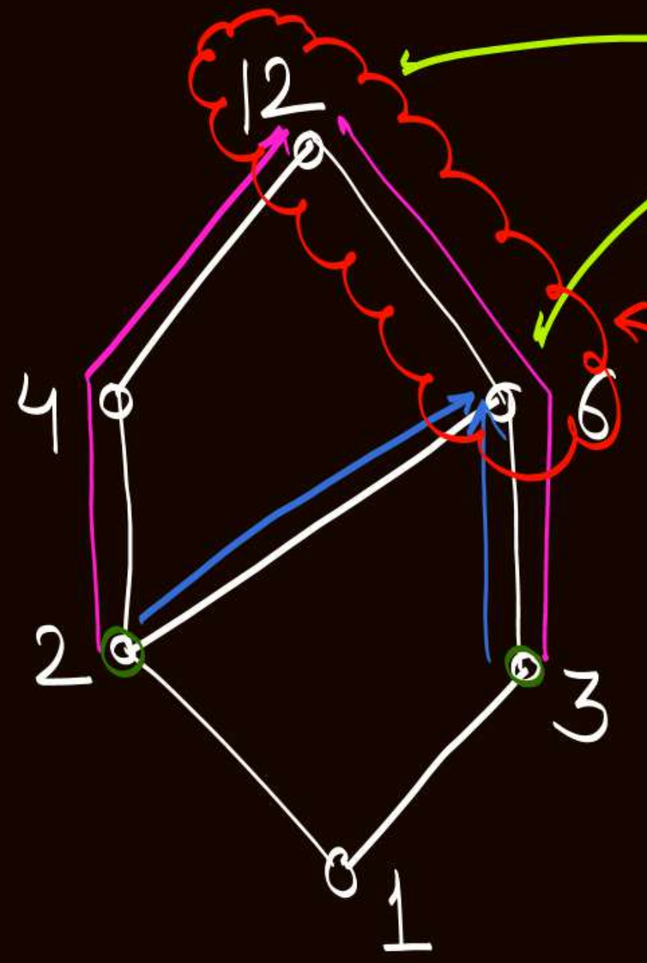
Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(D12, \div)$

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$



Both 12 & 6 are upper bounds of 2 & 3

Among 12 & 6, element 6 is at lower side

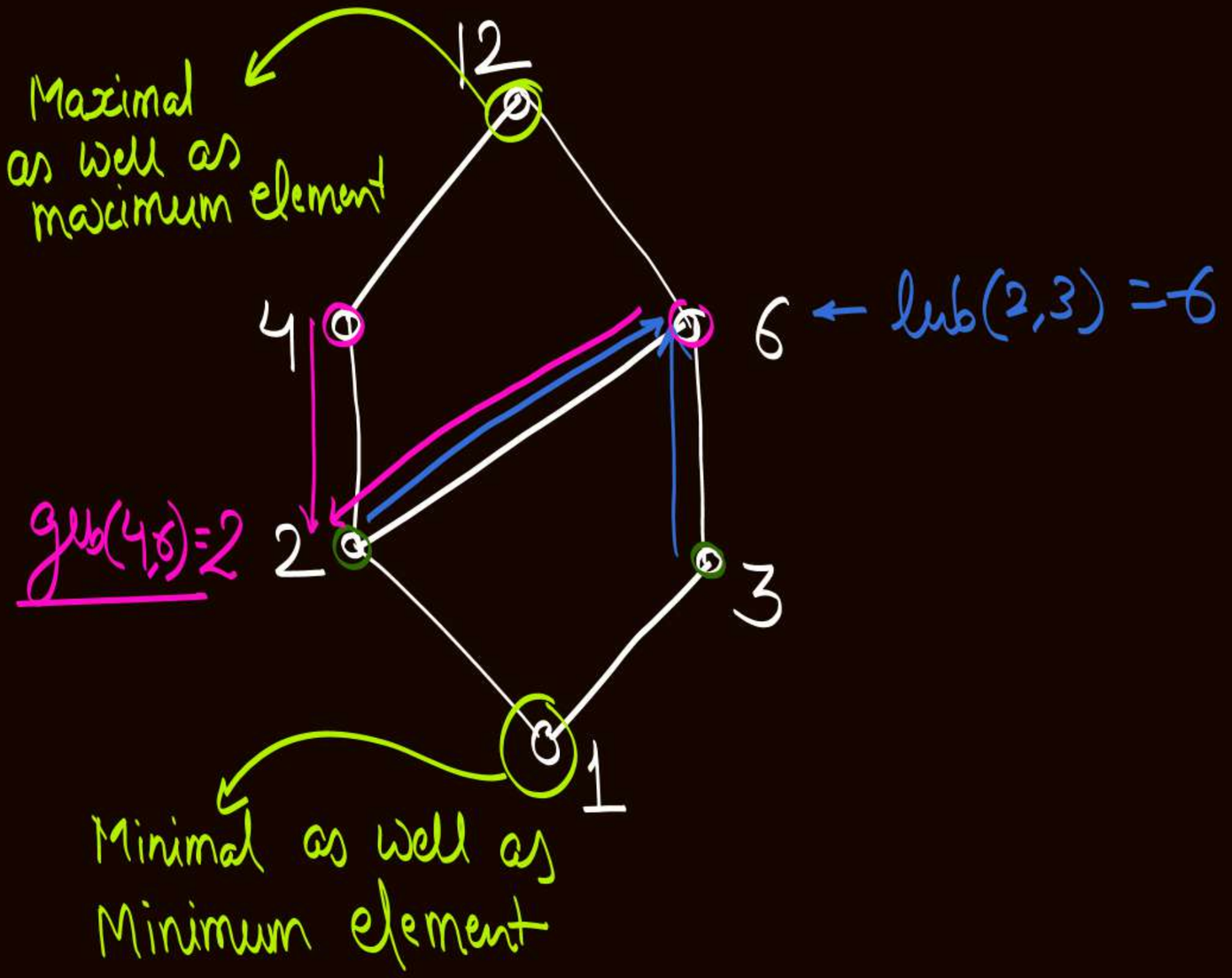
$\therefore lub(2,3) = 6$

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$

Join semi lattice as well as Meet semi lattice

$\therefore$ Lattice





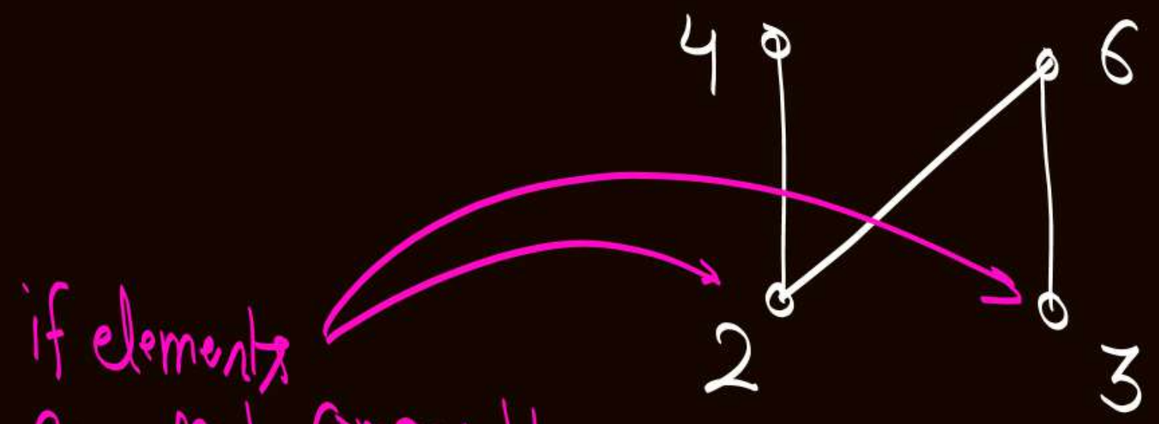
Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{2,3,4,6\},\div)$

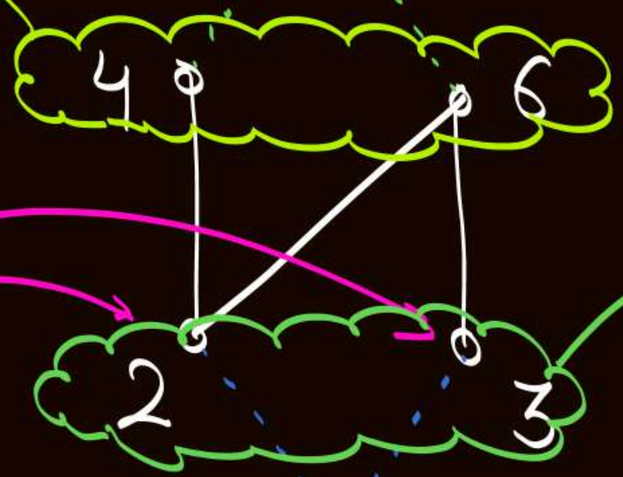
Slide

$(\{2, 3, 4, 6\}, \div)$



if elements are not comparable, then they may be at same level

$(\{2, 3, 4, 6\}, \div)$



- Both are Maximal elements

Two or more maximal
∴ No Maximum element

lub(4,6) = does not exist

∴ Not a Join Semi lattice

Both are Minimal elements

Two or more minimal
∴ No minimum element

glb(2,3) = does not exist

∴ Not a Meet Semi lattice

if elements are not comparable, then they may be at Same level

Neither a Join Semi lattice nor a Meet semi lattice

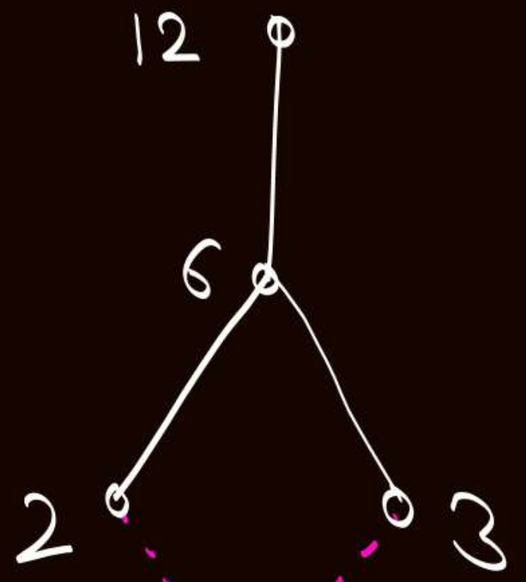


Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{2,3,6,12\},\div)$

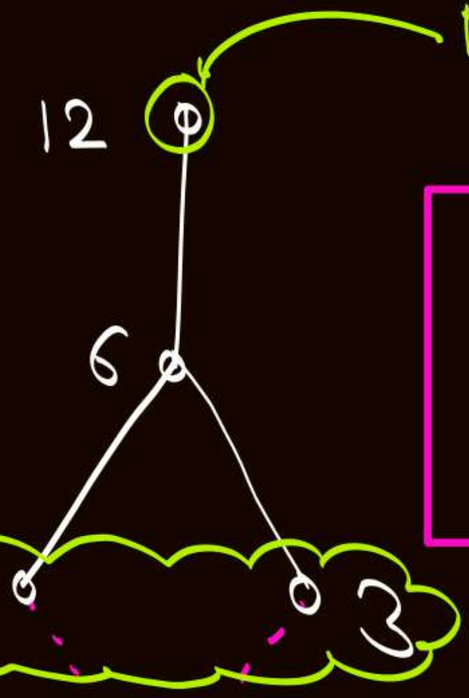
$(\{2, 3, 6, 12\}, \div)$



glb(2,3) = does not exist

$(\{2, 3, 6, 12\}, \div)$

It is a Join Semi lattice but, not a Meet Semi lattice



Maximal as well as Maximum element

lub exist for every pair of elements, ∴ It is a Join Semi lattice

Two minimal elements

∴ No minimum element

glb(2,3) = does not exist

∴ Not a Meet Semi lattice



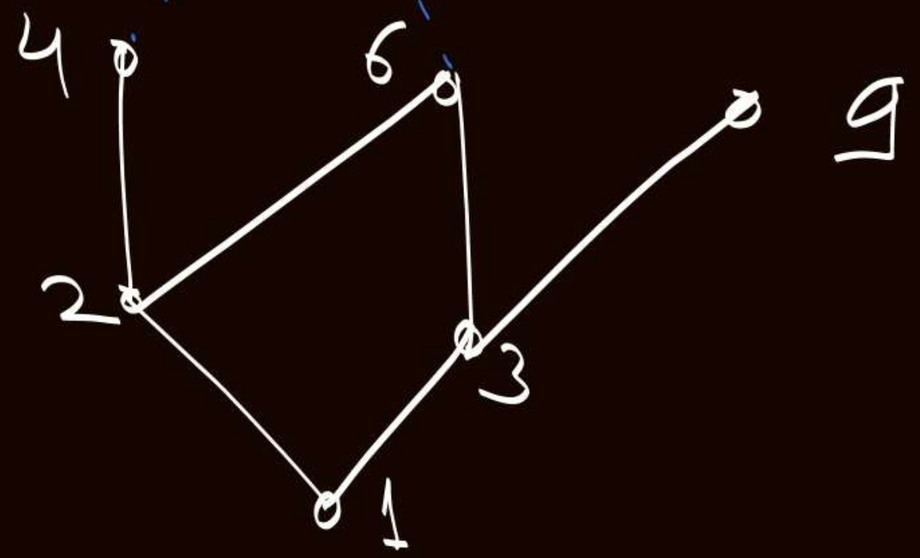
Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{1,2,3,4,6,9\},\div)$

$(\{1, 2, 3, 4, 6, 9\}, \div)$

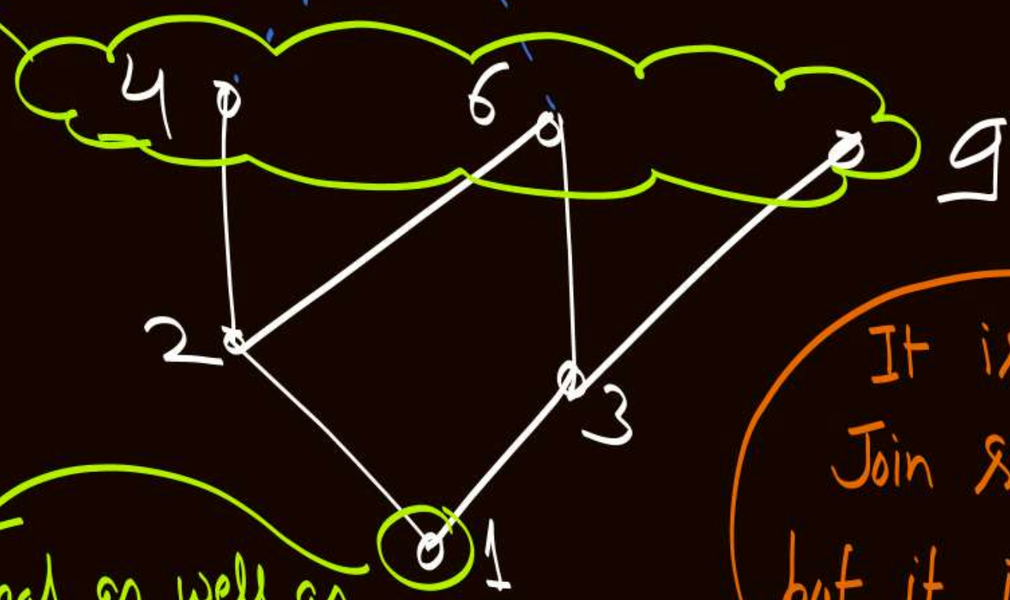
lub(4,6) - does not exist.



$(\{1, 2, 3, 4, 6, 9\}, \div)$

lub(4,6) does not exist.

∴ Not a Join Semi lattice



Three maximal elements

↓

∴ No maximum element

glb exist for every Pair of elements,

∴ It is a Meet Semi lattice

Minimal as well as Minimum element

It is not a Join Semi lattice, but it is a Meet Semi lattice

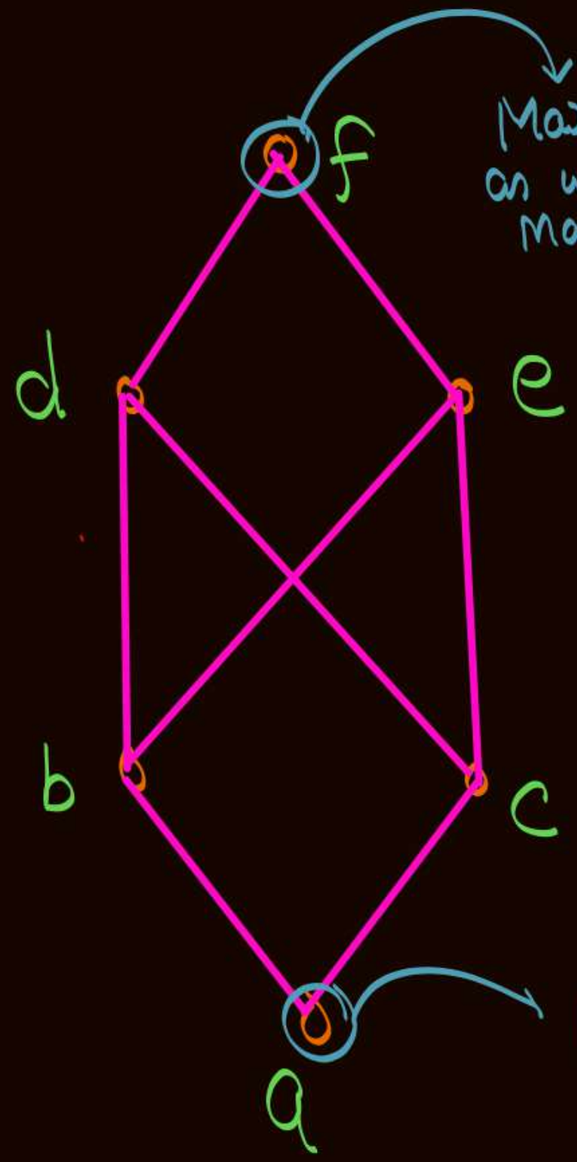
Note:-

① If there exist two or more maximal elements in a POSET, then it can not be a Join semi lattice.

② If there exist two or more minimal elements in a POSET, then it can not be a Meet semi-lattice

NOTE

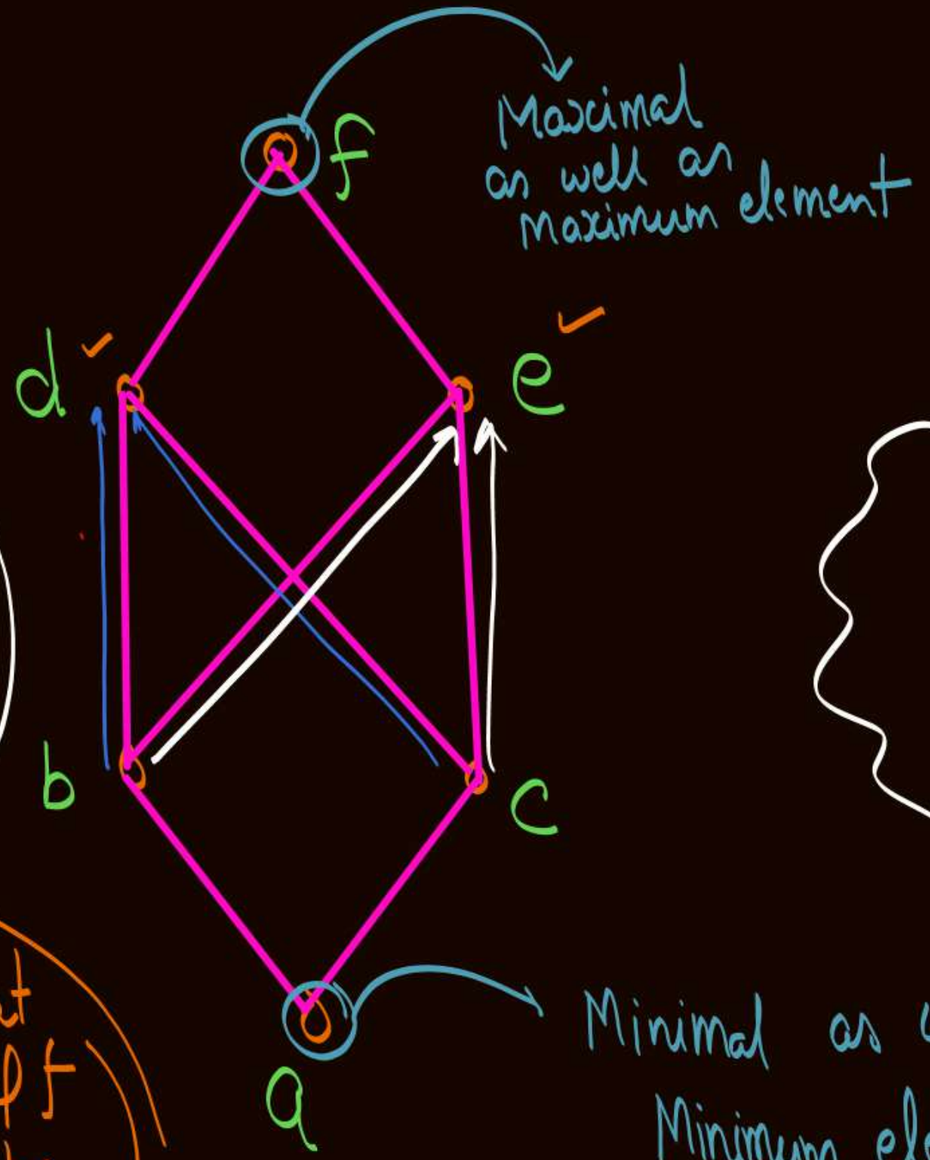
eg:



$R = \{ (a,a), (b,b), (c,c), (d,d), (e,e), (f,f)$
$(a,b), (a,d), (a,e), (a,f), (a,c)$
$(b,d), (b,e), (b,f),$
$(c,d), (c,e), (c,f),$
$(d,f), (e,f) \}$

NOTE

eg:



d, e, & f all are upper bounds of b & c

out of them d & e are at lower side of f ∴ f can not be least upper bound → lub of (b & c)

d & e are not Comparable, ∴ we can not decide the

We can not decide among 'd' & 'e'

lub(b, c) = does not exist

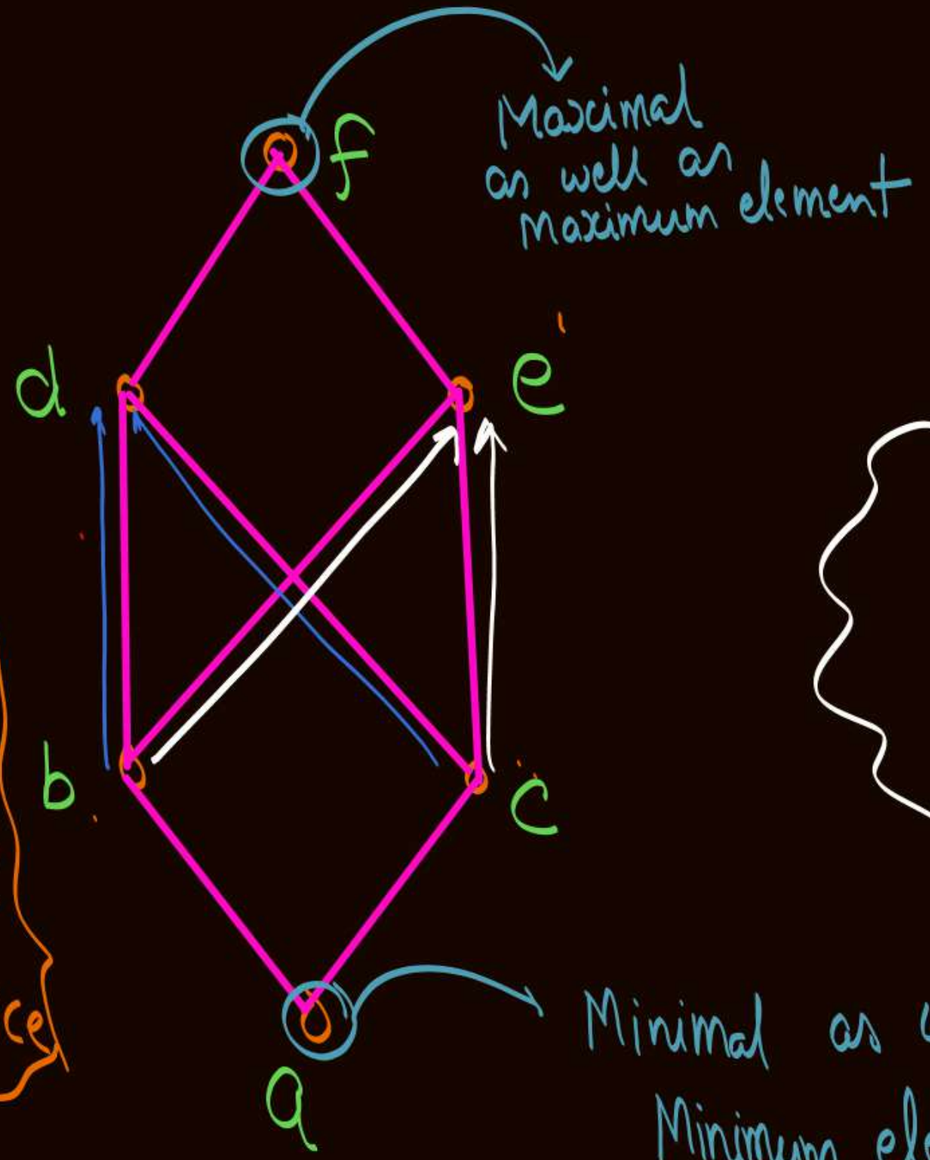
↳ In a lattice lub as well as glb should exist for every pair of elements & it must be unique

↳ ∴ given POSET is not a lattice

Similarly, glb(d, e) = does not exist.

NOTE

eg:



* Minimum as well as maximum element exist in this POSET, but this POSET is not a lattice

We can not decide among 'd' & 'e'

lub(b, c) = does not exist

↳ In a lattice lub as well as glb should exist for every pair of elements & it must be unique

↳ ∴ given POSET is not a lattice

Similarly, glb(d, e) = does not exist.



Topic : Sublattice

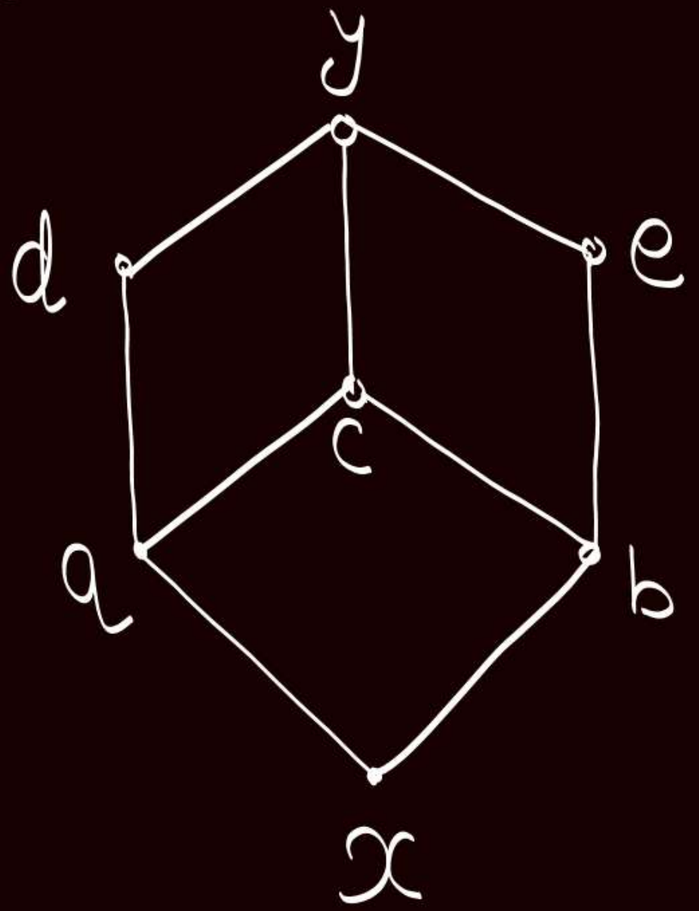
- Let $[L, \vee, \wedge]$ be a lattice, and M is a subset of set L.
- "M" w.r.t. same relation is called a sublattice of lattice L only if, (1) $[M, \vee, \wedge]$ must be a lattice {i.e. for every pair of elements of subset M lub & glb must exist in subset M.}

and (2) For every pair of elements of subset M, lub & glb must be same as lub & glb for that pair of elements in Lattice $[L, \vee, \wedge]$

i.e. $\forall a, b \in M$. $\underset{\text{in } M}{lub(a,b)} = \underset{\text{in } L}{lub(a,b)}$ & $\underset{\text{in } M}{glb(a,b)} = \underset{\text{in } L}{glb(a,b)}$

Slide

eg: Consider the following Lattice 'L'



$L = \{x, a, b, c, d, e, y\}$

Which of the following is/are sublattice of lattice L

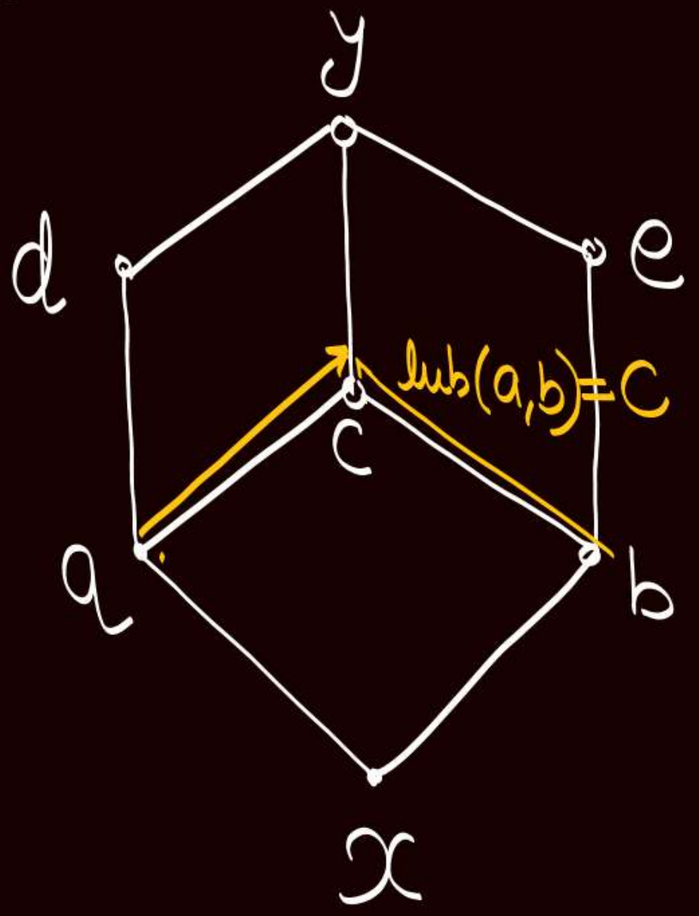
(a) $L_1 = \{x, a, b, y\}$

(b) $L_2 = \{x, a, c, y\}$

(c) $L_3 = \{x, c, d, y\}$

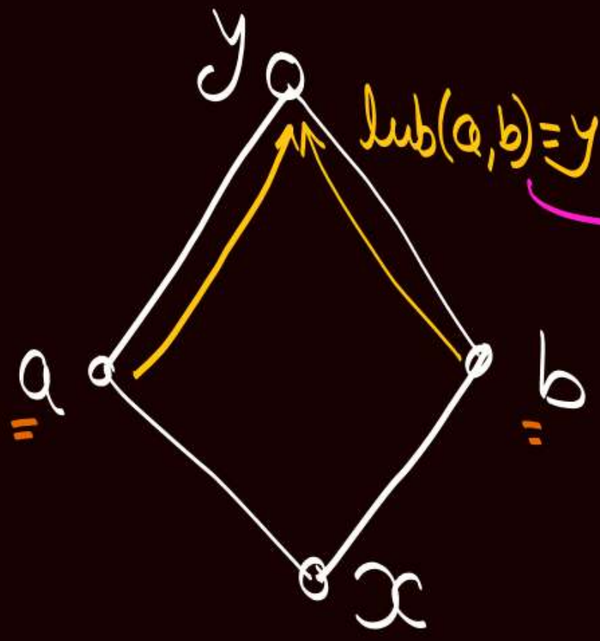
(d) $L_4 = \{x, a, e, y\}$

eg: Consider the following Lattice 'L'



$L = \{x, a, b, c, d, e, y\}$

(a) $L_1 = \{x, a, b, y\}$



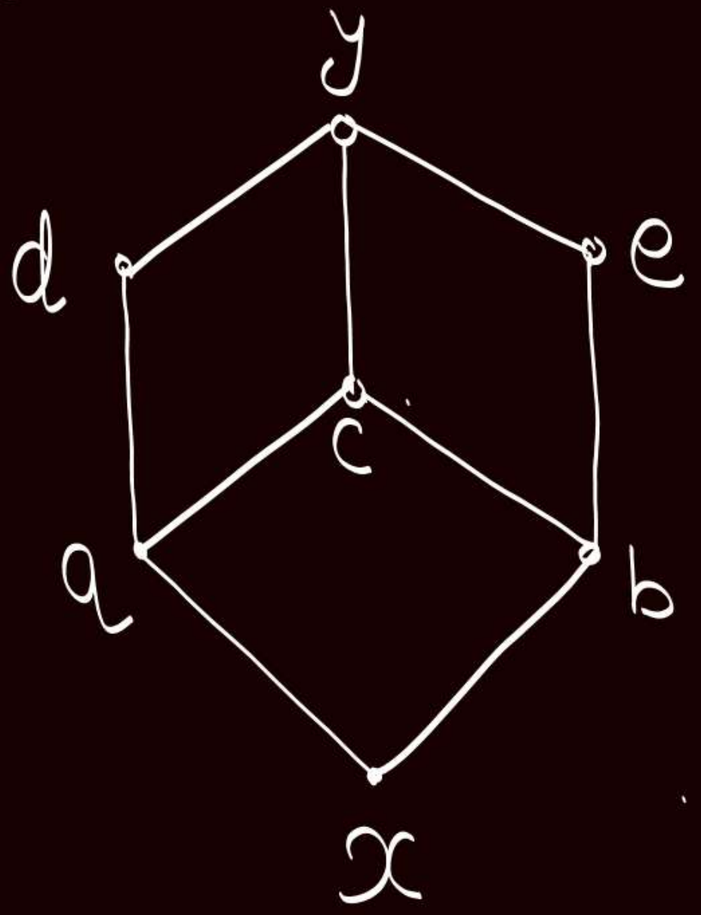
① L_1 is a lattice

Not same as lattice L

↓

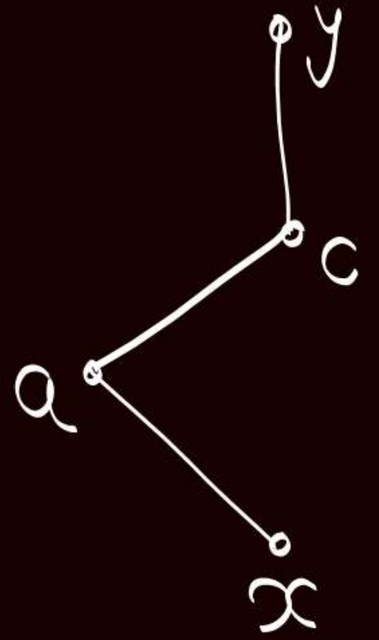
∴ L_1 is not a sublattice of lattice L

eg: Consider the following Lattice 'L'



$L = \{x, a, b, c, d, e, y\}$

(b) $L_2 = \{x, a, c, y\}$



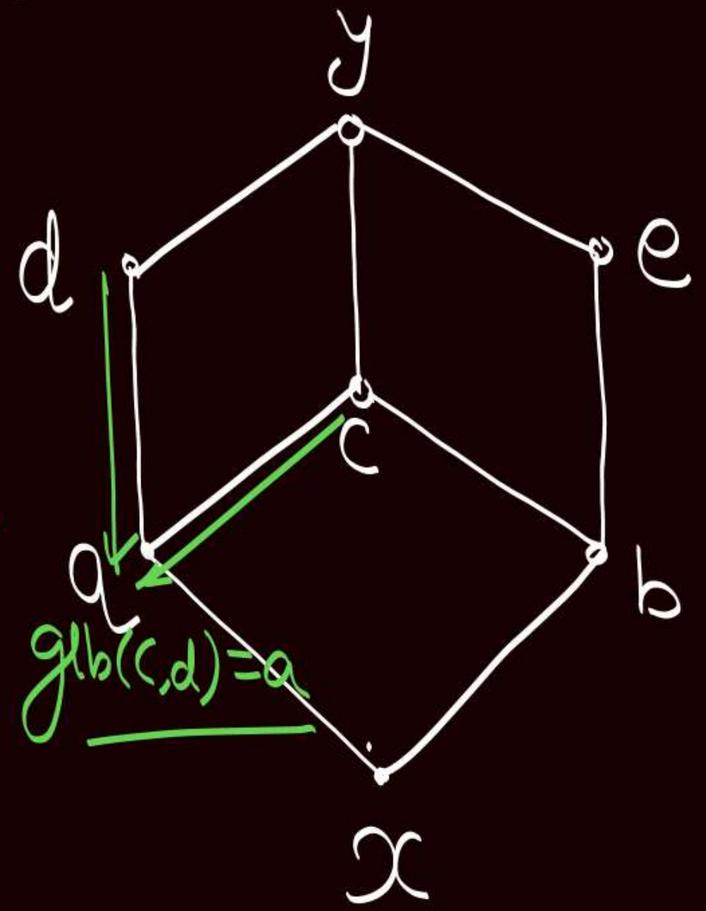
L_2 is a Subbattice of lattice L

① L_2 is a lattice

② x, a, c & y are in a line; in lattice L as well as in lattice L_2

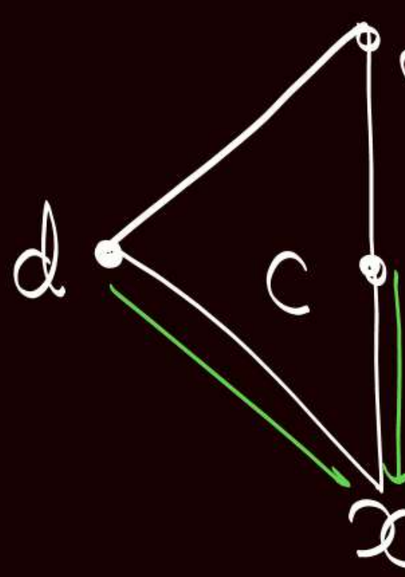
lub & glb for every Pair of elements in L_2 will be same as L

eg: Consider the following Lattice 'L'



$L = \{x, a, b, c, d, e, y\}$

(i) $L_3 = \{x, c, d, y\}$



① L_3 is a lattice

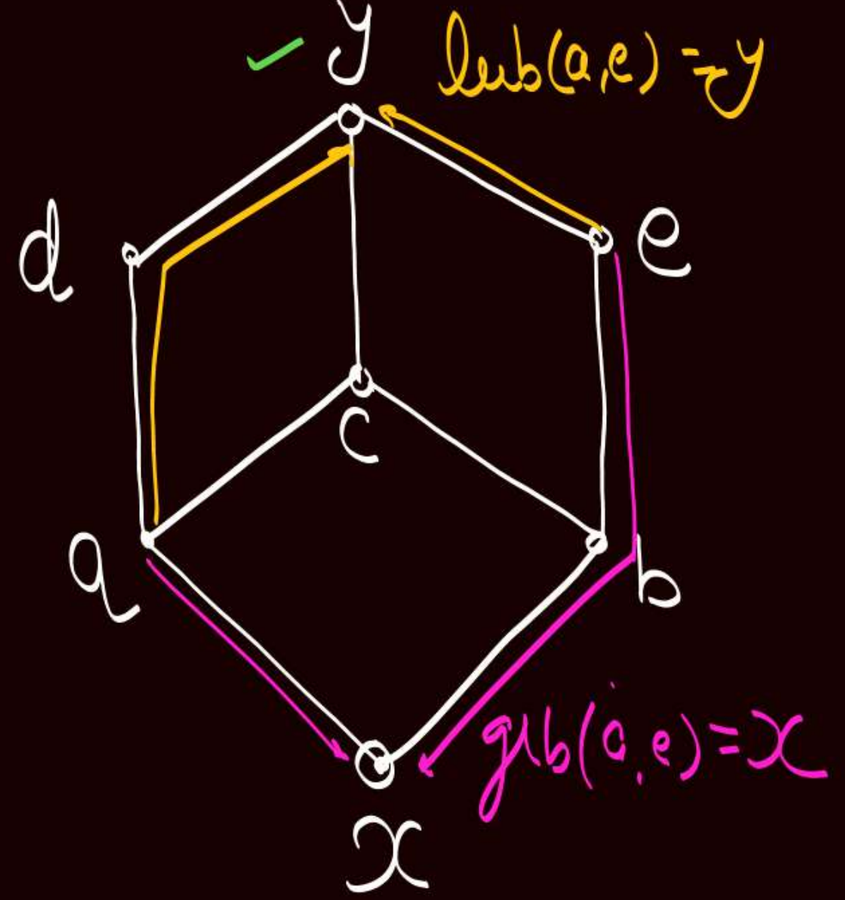
$glb(c,d) = a$

$glb(c,d) = x$

Not same as lattice L

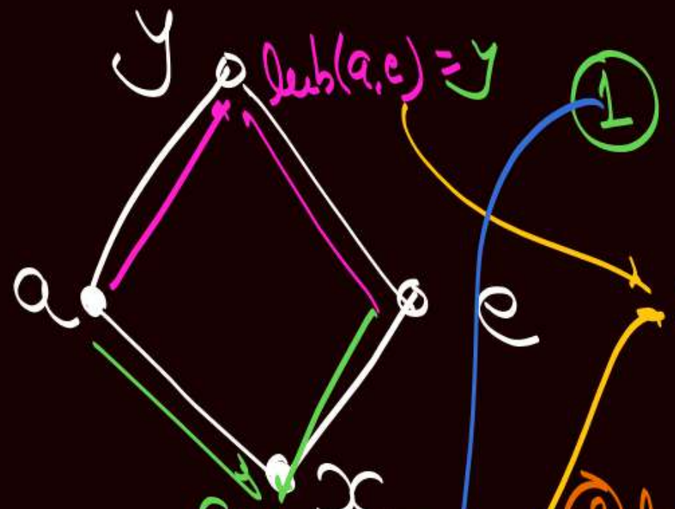
∴ L_3 is not a Sublattice of lattice L

eg: Consider the following Lattice 'L'



$L = \{x, a, b, c, d, e, y\}$

(d) $L_4 = \{x, a, e, y\}$



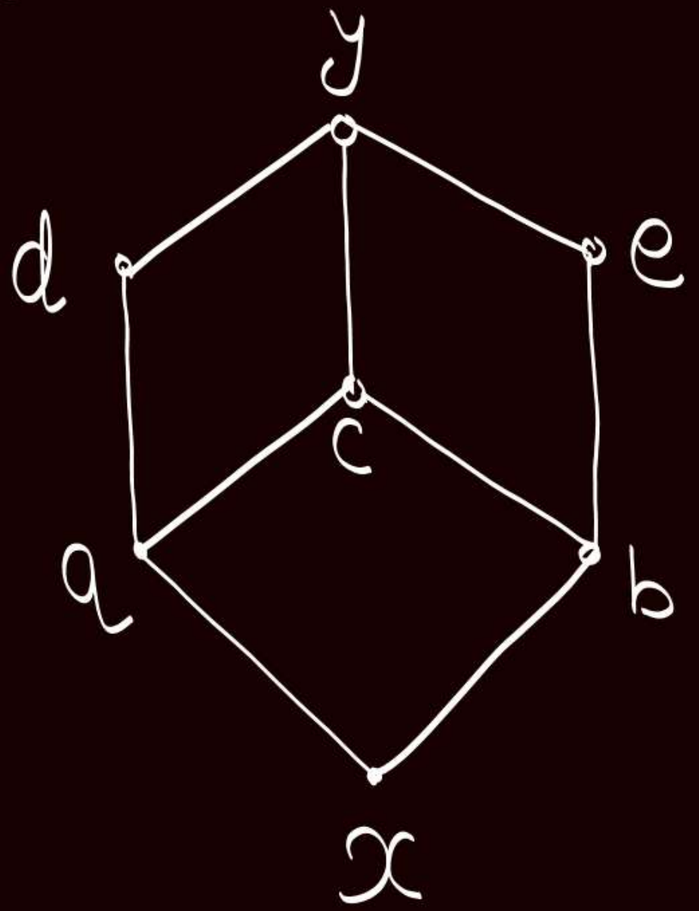
① L_4 is a lattice

same as L

② lub & glb for every pair of elements of L_4 is same as L

∴ L_4 is a sublattice of L

eg: Consider the following Lattice 'L'



$L = \{x, a, b, c, d, e, y\}$

Which of the following is/are sublattice of lattice L

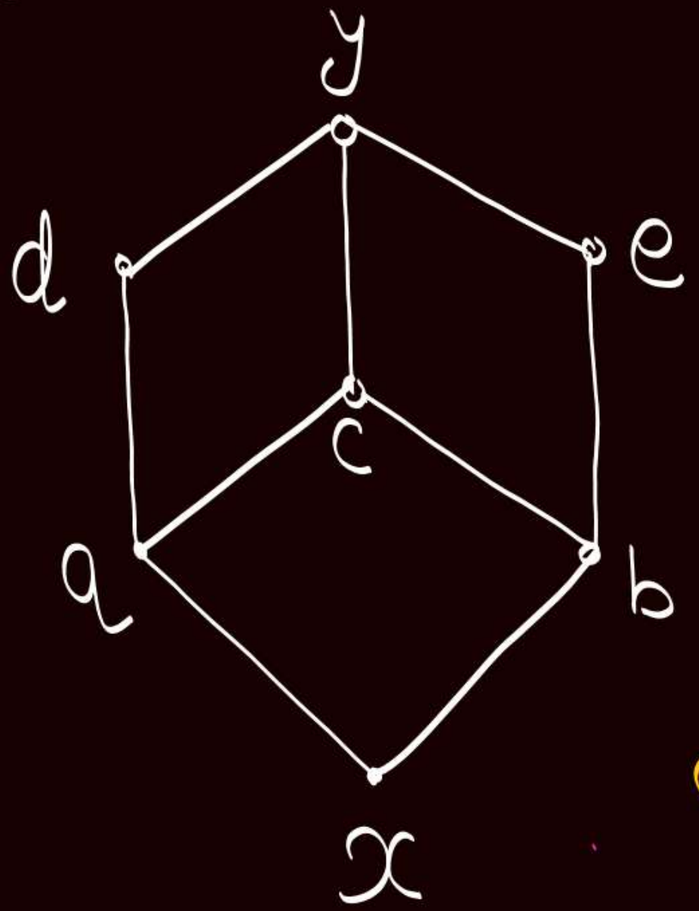
(a) $L_1 = \{x, a, b, y\}$

(b) $L_2 = \{x, a, c, y\}$

(c) $L_3 = \{x, c, d, y\}$

(d) $L_4 = \{x, a, e, y\}$

eg: Consider the following Lattice 'L'

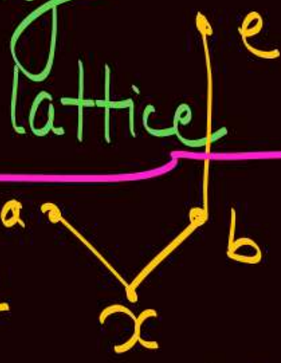


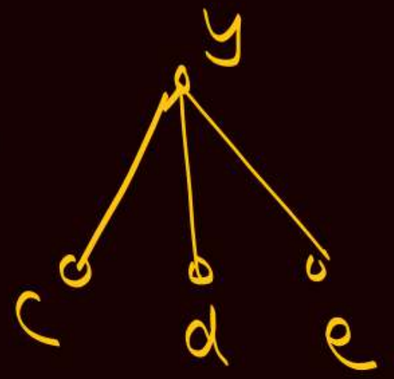
$L = \{x, a, b, c, d, e, y\}$

Which of the following is/are sublattice of lattice L

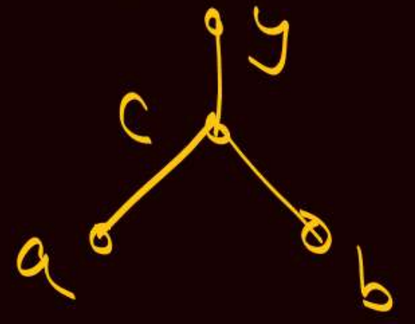
None of them

$L_5 = \{x, a, b, e\}$ =



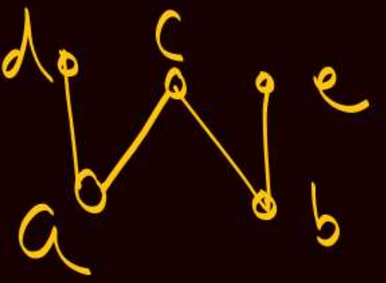


$L_6 = \{a, b, c, y\}$ =



$= L_7 = \{c, d, e, y\}$

$L_8 = \{a, b, c, d, e\}$ =





Topic : Bounded Lattice

Let $[L, \vee, \wedge]$ be a lattice

① If there exist an element $I \in L$.
such that

$$a \vee I = I, \quad \forall a \in L$$

$a \vee I$: lub(a, I)

then, I is called upper bound of lattice L
(or)
(Universal upper bound)

It is actually the maximum element of POSET wrt. lattice

Slide



Topic : Bounded Lattice

Let $[L, \vee, \wedge]$ be a lattice

② If these exist an element $O \in L$.
such that

$$a \wedge O = O, \quad \forall a \in L$$

$a \wedge O$ = glb(a, O)

then, O is called lower bound of lattice L
(or)
(Universal lower bound)

It is actually the minimum element of POSET, w.r.t lattice

Slide



Topic : Bounded Lattice

Let $[L, \vee, \wedge]$ be a lattice

If both I (universal upper bound) & O (universal lower bound) exist in lattice 'L', then L is called a bounded lattice

- I = Maximum (Greatest) element of POSET w.r.t. lattice
- O = Minimum (least) element of POSET w.r.t. lattice

Slide



Topic : Bounded Lattice

Let $[L, \vee, \wedge]$ be a lattice

If both Minimum as well as Maximum element exist in lattice, then 'L' is a bounded lattice.

otherwise, unbounded lattice

Slide

Note ① There are some lattice which are not bounded.

eg: $(\mathbb{N}, \leq)$ is a lattice, but it is not a bounded lattice

Set of natural No.

POSET diagram ⇒



'1' is the universal lower bound

but universal upper bound does not exist.

eg $(Z, \leq)$ is also a lattice, but not bounded

Set of all integers

I = Universal upper bound = does not exist

O = Universal lower bound = does not exist.

Note: ② If lattice $[L, \vee, \wedge]$ is an unbounded lattice, then underlying set 'L' is an infinite set, but Converse of the statement need not be true

eg: let $A = \{x \mid x \in R \text{ and } 0 \leq x \leq 1\}$

it is an infinite set

but $(A, \leq)$ is a bounded lattice

where I = universal upper bound = '1'

0 = universal lower bound = '0'

Note:- ③ Every finite lattice is a bounded lattice.



Topic : Complement of an element in a lattice

Note:- Complement of an element in a lattice can be defined only if lattice is a bounded lattice.

{i.e.; if lattice is not a bounded lattice then Complement does not exist for any element of that lattice}

Let $[L, \vee, \wedge]$ be a bounded lattice

where, I = universal upper bound & O = universal lower bound

for an element $a \in L$ if there exist any element $b \in L$

such that (i) $a \vee b = I$ {i.e. lub(a,b) = maximum element}

& (ii) $a \wedge b = O$ {i.e. glb(a,b) = Minimum element}

then 'a' & 'b' are called Complement of each other

Slide



Topic : Complement of an element in a lattice

① In a bounded lattice,
'I' & '0' are always Complement of each other
The only Complement of element I = 0
& only Complement of element 0 = I

② In a bounded lattice,
Complement need not exist for every element of lattice;
and if exist for any element then it need not be unique
i.e. In a bounded lattice number of Complements of an element can be 0 or more

Slide

Notes:

① A totally ordered set is always a lattice.

② A totally ordered set may or may not be a bounded lattice,

Both are totally ordered set

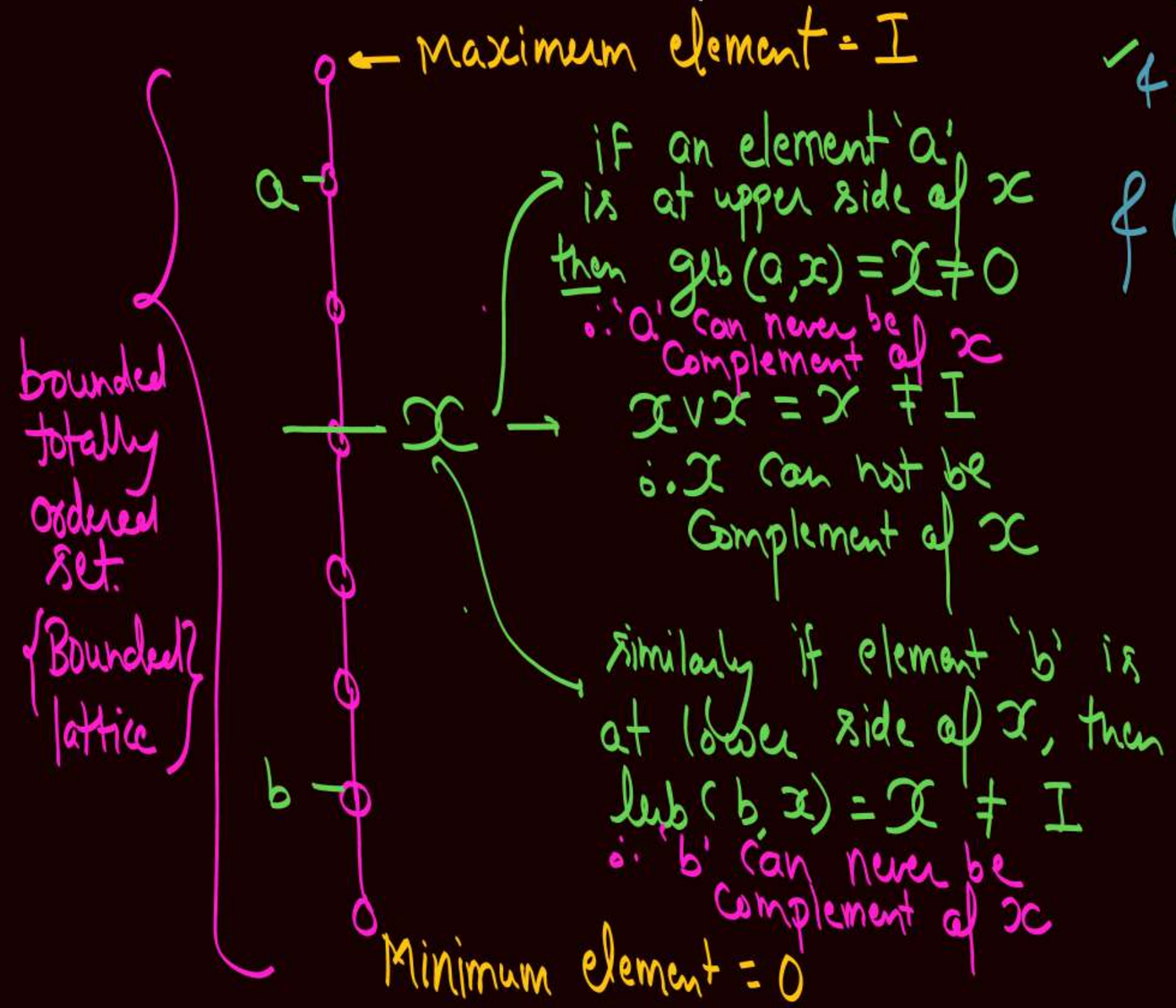
eg: $(\{1,2,3,4,5\}, \leq)$ is a bounded lattice

eg: $(\mathbb{N}, \leq)$ is an unbounded lattice.

③ If totally ordered set is an unbounded lattice then Complement does not exist for any element of that lattice

Note: ④ If totally ordered set is bounded lattice, then

- (i) Complement of $I = 0$
- (ii) Complement of $0 = I$
- (iii) Complement does not exist for any other element of totally ordered set

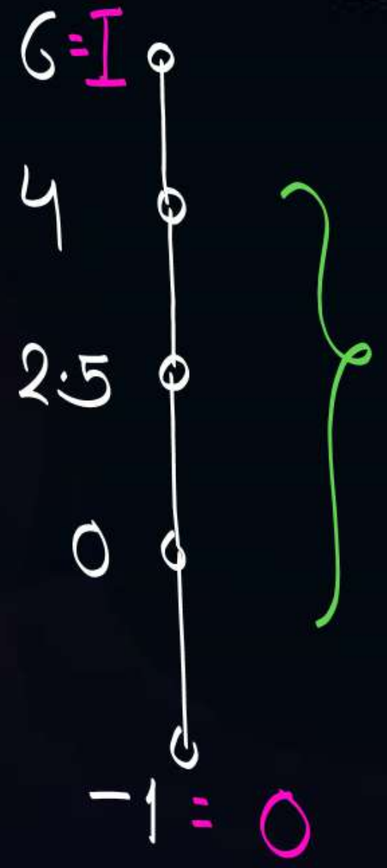


Note: Two elements on the same line in the Hasse diagram can never be complement of each other, except for minimum & maximum elements

Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSET

$(\{-1,0,2.5,4,6\}, \leq)$



$\overline{-1} = 6$ & $\overline{6} = -1$

$\overline{0}$ = does not exist

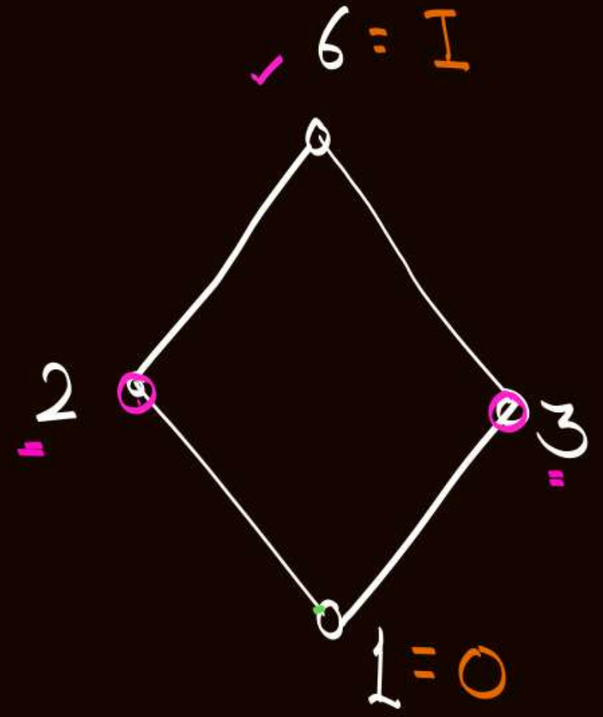
$\overline{2.5}$ = does not exist

$\overline{4}$ = does not exist

Slide

$(D_6, \div)$

$D_6 = \{1, 2, 3, 6\}$



$\bar{I} = 0$ & $\bar{0} = I$

$\bar{6} = 1$ & $\bar{1} = 6$

$2 \vee 3 = 6 = I$

$2 \wedge 3 = 1 = 0$

$\therefore$ 2 & 3 are Complement of each other

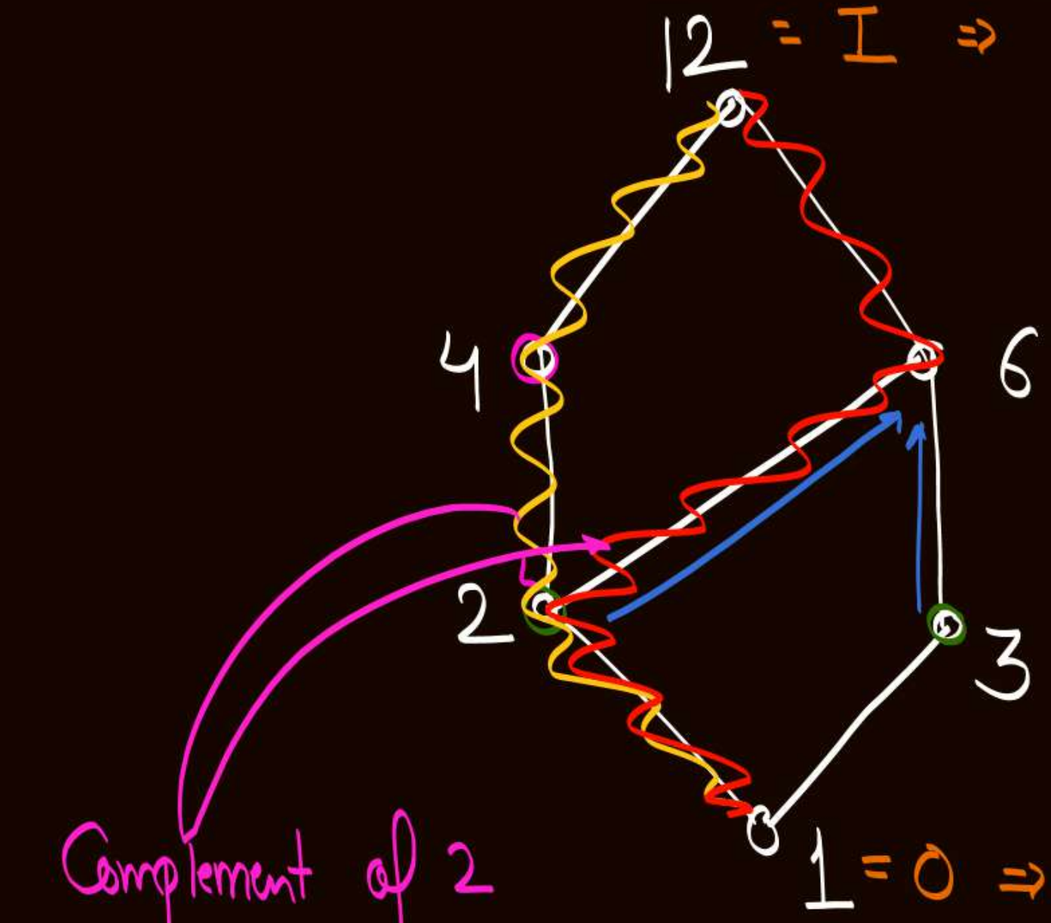
i.e. $\bar{2} = 3$

& $\bar{3} = 2$

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$

$\overline{2}$ = does not exist



$12 = I \Rightarrow \overline{12} = 1$

$1 = 0 \Rightarrow \overline{1} = 12$

Complement of 2 can never exist on this yellow color line or Red color line

$\Rightarrow$ ∴ only possibility = 3

$2 \vee 3 = 6 \neq I$

∴ '3' is not Complement of 2

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$

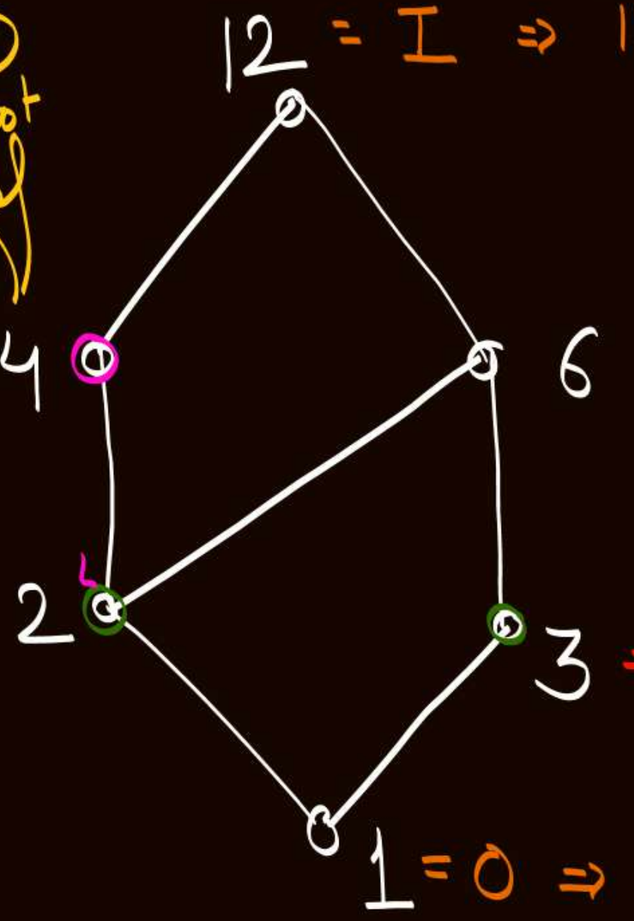
$\bar{2}$ = does not exist

$\bar{3} = 4$

$\bar{4} = 3$

$\bar{6}$ = does not exist

$glb(4,6) = 2 \neq 0$
$\therefore$ 4 & 6 can not be complement of each other

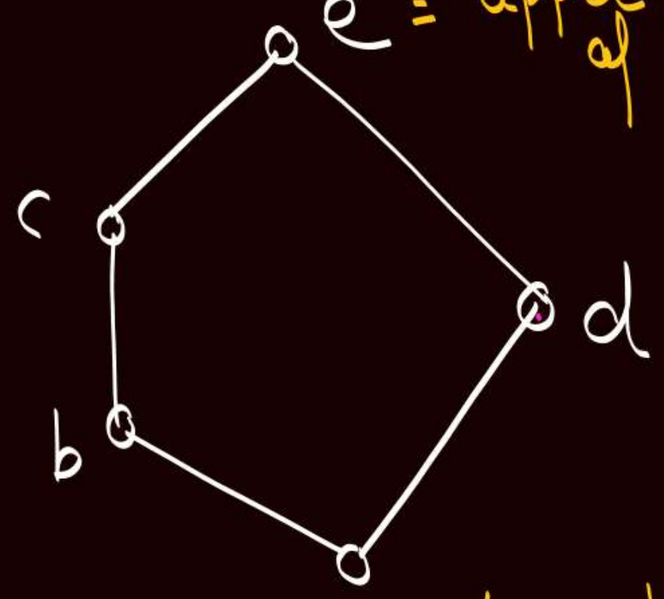


$12 = I \Rightarrow \bar{12} = 1$

$1 = 0 \Rightarrow \bar{1} = 12$

$3 \vee 4 = 12 = I$
$3 \wedge 4 = 1 = 0$
$\therefore$ 3 & 4 Complement of each other

eg. Find the complements of every element in the following lattice.



e = upper bound of lattice

a = lower bound of lattice

$\bar{a} = e$

$\bar{e} = a$

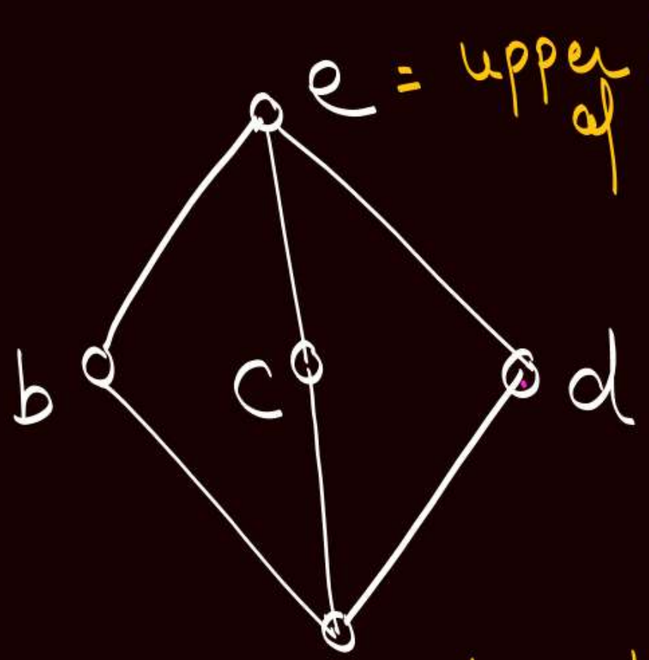
$\bar{b} = d$

$\bar{c} = d$

$\bar{d} = b, c$

some elements may have more than one complement

eg. Find the complements of every element in the following lattice.



$\bar{a} = e$

$\bar{e} = a$

$\bar{b} = c, d$

$\bar{c} = b, d$

$\bar{d} = b, c$



Topic : Complemented lattice

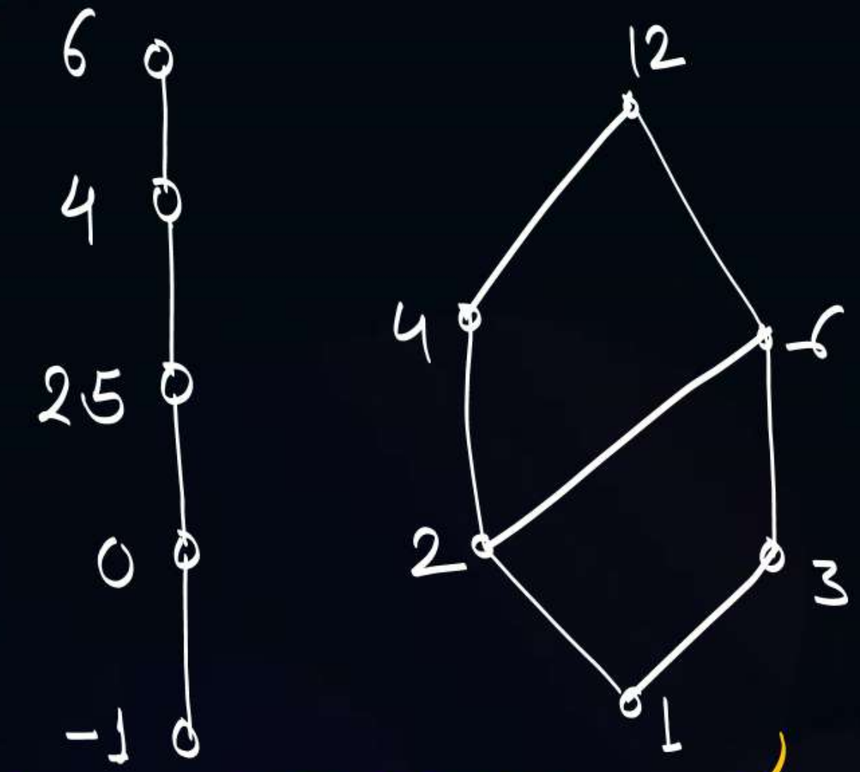
A bounded lattice in which Complement exist for every element of the lattice is called Complemented lattice

- In a complemented lattice, Complement must exist for every element but it need not be unique

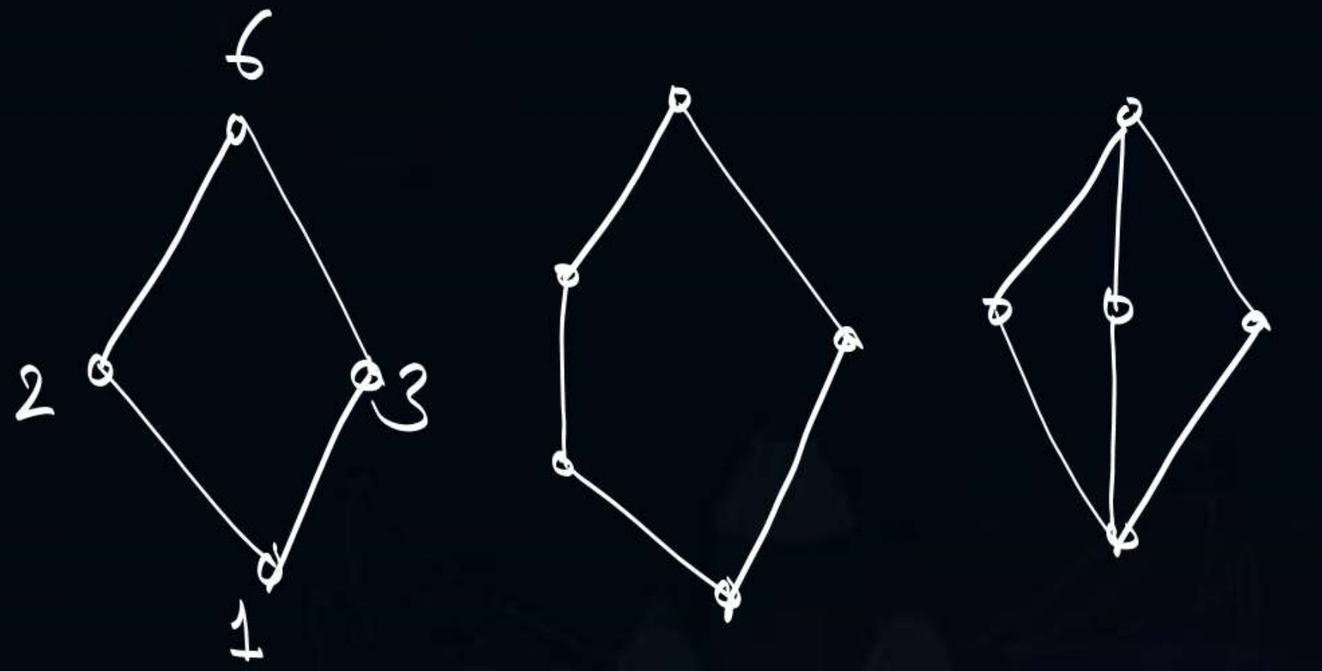
  i.e; in a complemented lattice every element has at least one complement

Slide

Topic : Complemented lattice



Not Complemented lattice



Complemented lattice

Slide



Topic : Distributive lattice

A lattice $[L, \vee, \wedge]$ is called a distributive lattice if and only if following Poopeties holds tore

(i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

& (ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

$\Rightarrow \forall a, b, c \in L$

Slide



Topic : Distributive lattice

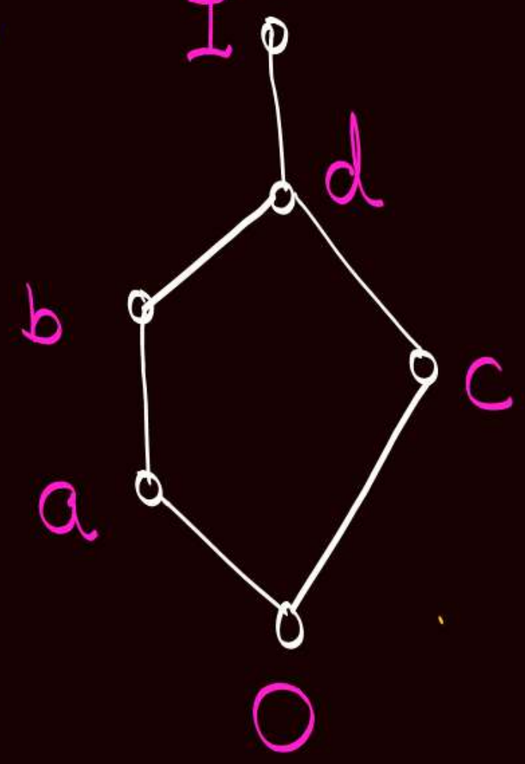
In a distributive lattice complement of an element need not exist, but if exist then it must be unique

i.e. in a distributive lattice every element has at most one complement

- If, lattice is a distributive lattice, then every element has at most one complement {Converse of the statement need not be true}
  {If every element has at most one complement, then lattice may or may not be distributive.}
- If, any element of the lattice has two or more complements then that lattice is not a distributive lattice

Slide

L ⇒



We know $a, b, c \in L$

i. following must be true for distributive lattice

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

but ⇒ $a \vee O$ $\quad b \wedge d$

⇒ a $\quad \neq \quad b$

↳ i.e. distributive property does not hold true hence given lattice is not a distributive lattice

$\bar{O} = I$ & $\bar{I} = O$

$\bar{a}$ = does not exist

$\bar{b}$ = does not exist

$\bar{c}$ = does not exist

$\bar{d}$ = does not exist

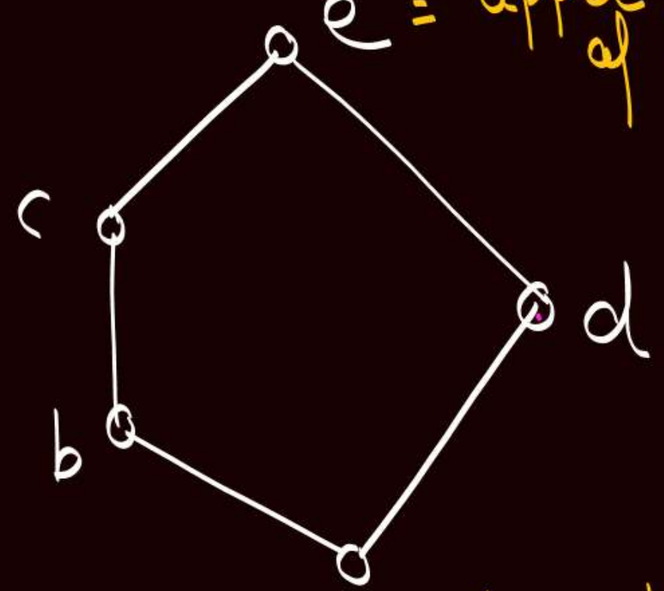
} Every element has 0 or 1 complement

Note:- ① If we choose three elements on the same line, then they will always satisfy distributive property.

② In a totally ordered set any triple of elements will always lie on the same line,

∴ A lattice w.r.t. TOSET is always distributive

eg. Find the complements of every element in the following lattice.



$\bar{a} = e$

$\bar{e} = a$

$\bar{b} = d$

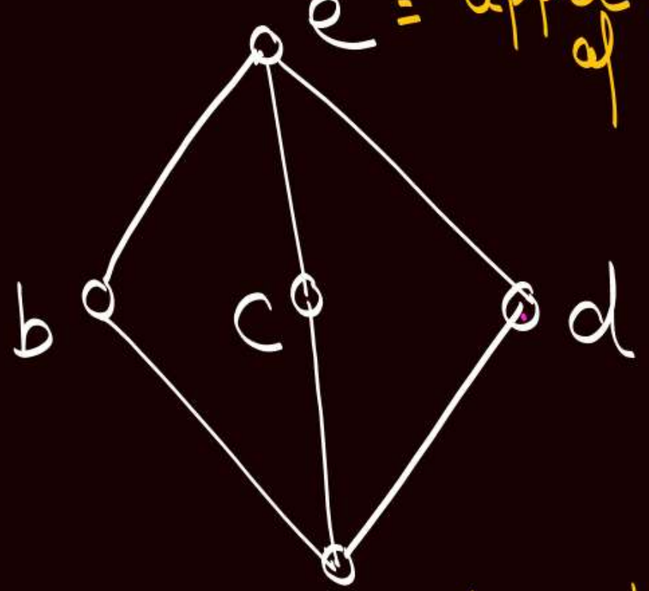
$\bar{c} = d$

$\bar{d} = b, c$

Some elements have more than one complements.

∴ Not a distributive lattice

eg. Find the complements of every element in the following lattice.



$\bar{a} = e$

$\bar{e} = a$

$\bar{b} = c, d$

$\bar{c} = b, d$

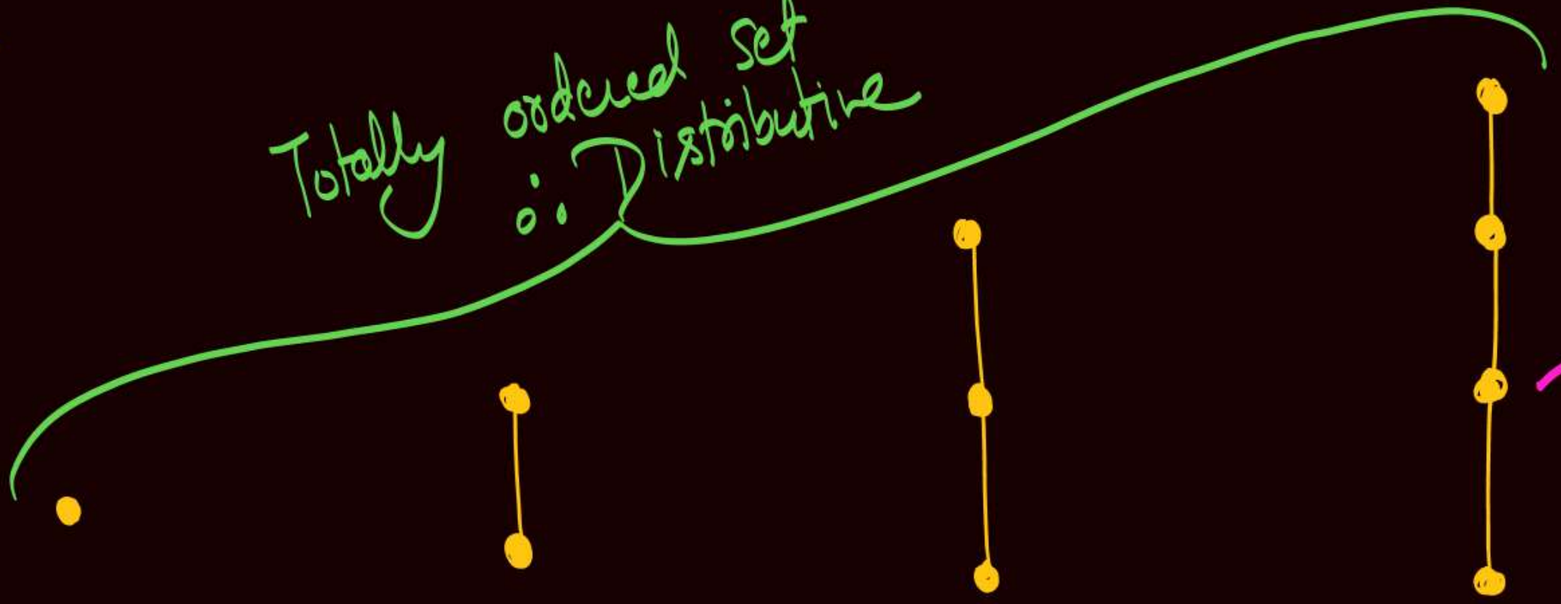
$\bar{d} = b, c$

Some elements have more than one complement.

∴ Not a distributive lattice

Note

Totally ordered set
∴ Distributive



It is also distributive

Hasse diagram w.r.t. lattice of single element

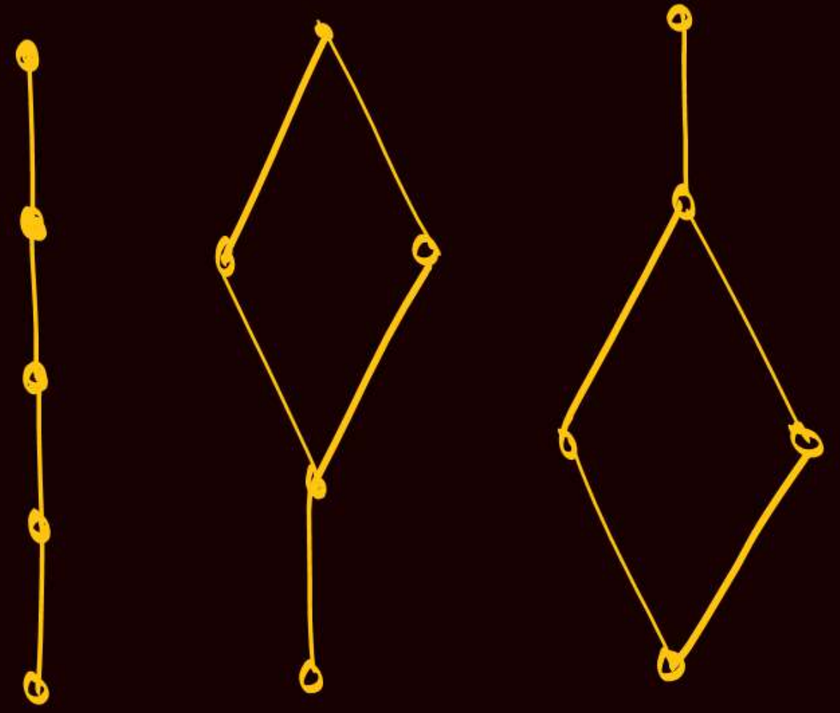
Hasse diagram w.r.t. lattice of two elements

It is the only Hasse diagram Possible w.r.t lattice of three elements

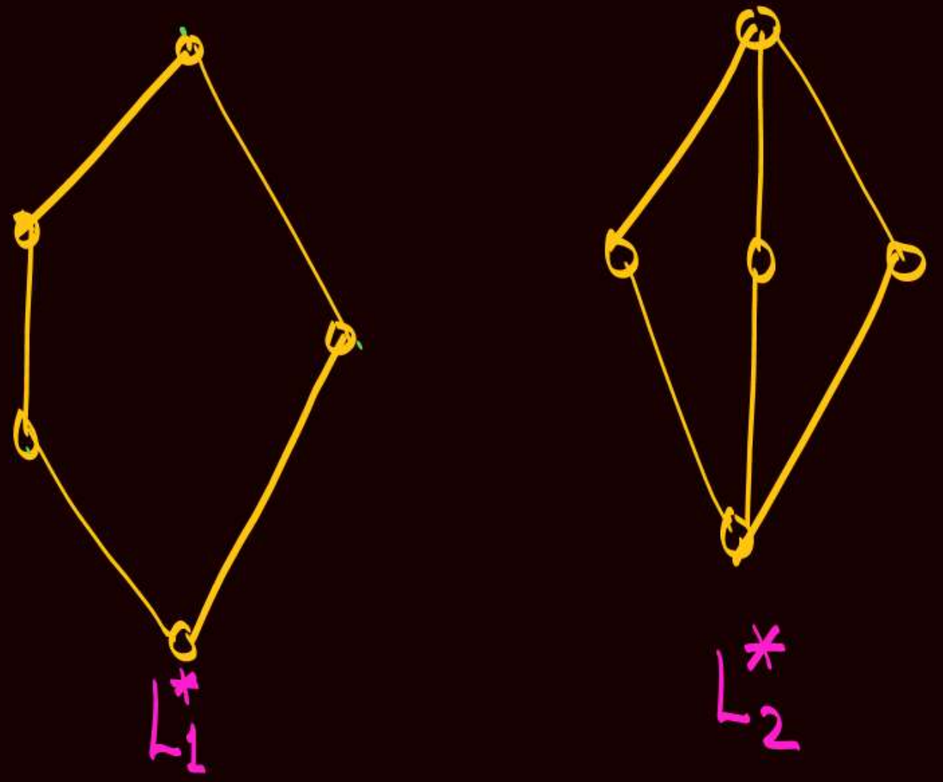
This are the only two Hasse diagrams Possible w.r.t. lattices of four elements

Note: Any lattice with 4 or less elements is always distributive

Note Hasse diagrams possible w.r.t lattices of 5 elements are,



Lattices w.r.t. above three Hasse diagram will always be distributive

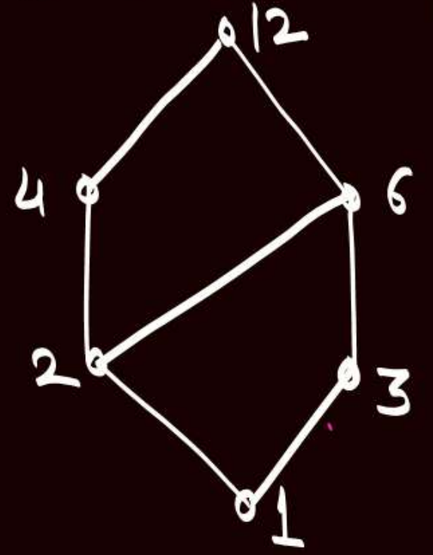


L_1 & L_2 are the only two lattices with 5 elements which are not distributive

Note:- A lattice is not distributive lattice if and only if it has a sub-lattice which is isomorphic {similar} to L_1^* or L_2^*

Q: Check whether the following lattice is a distributive lattice or not.

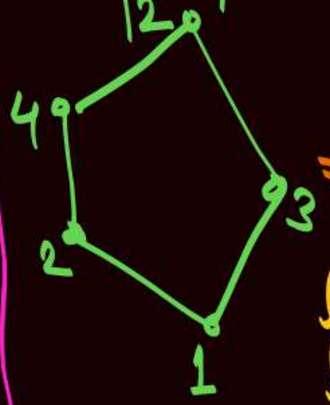
$(D_{12}, \div)$



"L"

There is no subset of L w.r.t. which the Hasse diagram can be isomorphic to L_2^*

Subset $\{1, 2, 3, 4, 12\}$ of L has a hasse diagram which is isomorphic to L_1^*



$\Rightarrow$ but resulting lattice is not a sub-lattice of L

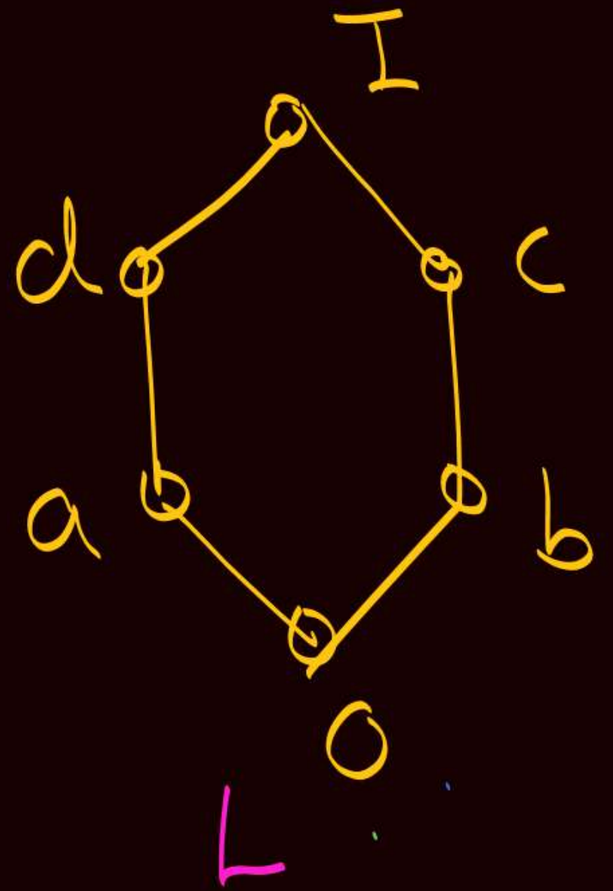
$lub(2,3) = 12$ in this lattice
$lub(2,3) = 6$ in lattice L

There is no other subset of L w.r.t. which the Hasse diagram will be isomorphic to L_1^* or L_2^*

i.e, there is no sublattice of L which is isomorphic to L_1^* or L_2^*

$\therefore$ L is a distributive lattice

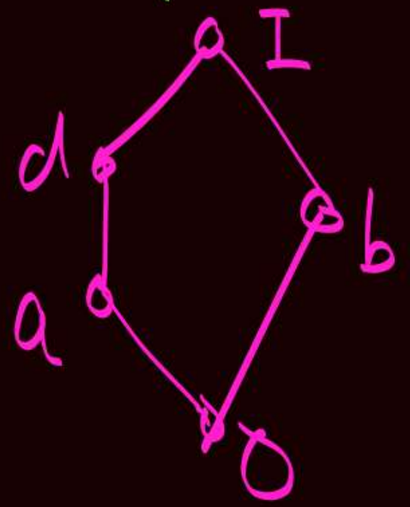
eg: Check whether the following lattice is distributive or not



$\bar{a} = b, c$

$\therefore$ Not distributive

$\Rightarrow$ It has a subset $\{O, I, a, b, d\}$ s.t

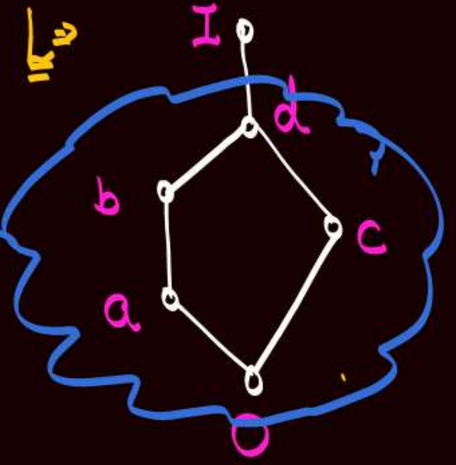


is isomorphic to L_1*

and it is sublattice of L

$\therefore$ L is not a distributive lattice

Another approach,



$\bar{O} = I$ & $\bar{I} = O$

$\bar{a}$ = does not exist

$\bar{b}$ = does not exist

$\bar{c}$ = does not exist

$\bar{d}$ = does not exist

} Every element has 0 or 1 complement

We know $a, b, c \in L$

∴ following must be true for distributive lattice

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

but ⇒ $a \vee 0 \qquad b \wedge d$

⇒ $a \neq b$

↳ i.e. distributive property does not hold true

hence given lattice is not a distributive lattice

it has a sublattice which is isomorphic to L_1^*

∴ Not a distributive lattice



Topic : Boolean lattice

A lattice which is both Complemented as well as distributive is called Boolean algebra or Boolean lattice

i.e. at least one complement

i.e. at most one complement

- A lattice is called a boolean lattice if and only if every element has a unique (Exactly 1) Complement

Slide

H.W

Which of the following statements is/ are not true

a) If A is any finite set then $[P(A), \subseteq]$ is distributive lattice
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice
d) Every totally ordered set is bounded
e) Every distributive lattice is bounded
f) Every distributive lattice is a complemented lattice

Slide

H.W.

Which of the following lattice is /are not distributive?

(A) $(D_{125}, \div)$
(B) $(P(A), \subseteq)$
(C) $(D_{12}, \div)$
(D) $(\{1, 2,3,5,30\}, \div)$
(E) $(\{1, 2,3, 4,12\}, \div)$

Slide

H.W.

Which of the following lattice is /are Boolean Algebra?

(A) $(D_{125}, \div)$
(B) $(P(A), \subseteq)$
(C) $(D_{12}, \div)$
(D) $(\{1, 2,3,5,30\}, \div)$
(E) $(\{1, 2,3, 4,12\}, \div)$

Slide



2 mins Summary

Topic Complemented lattice

Topic Distributive lattice

Topic Boolean lattice / Boolean algebra

THANK - YOU